

Concept 1 | Binomial and Multinomial Theorems

English Student Edition • Focused formulas and guided practice

Formula opener

Binomial

$$\binom{n}{r} a^r b^{n-r}$$

Multinomial

$$\binom{n}{r} \binom{n-r}{s} a^r b^s c^{n-r-s}$$

Exponent check

$$r + (n - r) = n$$

Worked Solutions

Binomial and Multinomial Theorems

Q001 Challenge

For $(2x - 3)^5$, find the coefficient of x^3 and the sum of all coefficients without writing the full expansion:

Solution

1. Use the binomial term rule: $q = 3$ and $r = 2$, so the coefficient is 5C_2 times 72.
2. For the sum of coefficients, evaluate the expression at $x = 1$: -1 .

Correct option: A

Final answer

Coefficient = 720; sum = -1.

Worked Solutions

Binomial and Multinomial Theorems

Q002 Challenge

Find the coefficient of x^2y^3z in $(x + y + z)^6$, then find the coefficient of x^2yz^3 in $(x - 2y + z)^6$:

Solution

1. The first exponents are (2,3,1), so use $6!/(2!3!1!) = 60$.
2. The second exponents are (2,1,3); the factor (-2) is raised to the first power of y, giving -120.

Correct option: A

Final answer

First coefficient = 60; second coefficient = -120.

Worked Solutions

Binomial and Multinomial Theorems

Q003 Challenge

For $(x^2 + x + 1)^4$, find the greatest coefficient and verify it by identifying the coefficient of x^4 :

Solution

1. Expand and collect like powers: the central coefficient is 19.
2. The coefficient of x^4 is also 19; it is therefore the greatest.

Correct option: A

Final answer

Greatest coefficient = 19.

Worked Solutions

Binomial and Multinomial Theorems

Q004 **Written****Expand $(x + 2)^5$ using the Binomial Theorem.****Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^5 = x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$.

Final answer

$$x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$$

Q005 **MCQ****Expand $(3x - 2)^4$ using the Binomial Theorem:****Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x - 2)^4 = 81x^4 - 216x^3 + 216x^2 - 96x + 16$.

Correct option: D**Final answer**

$$81x^4 - 216x^3 + 216x^2 - 96x + 16$$

Worked Solutions

Binomial and Multinomial Theorems

Q006 Written

Expand $(x + 2)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^6 = x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$.

Final answer

$$x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$$

Q007 MCQ

Expand $(x + 1)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 1)^4 = x^4 + 4x^3 + 6x^2 + 4x + 1$.

Correct option: B**Final answer**

$$x^4 + 4x^3 + 6x^2 + 4x + 1$$

Worked Solutions

Binomial and Multinomial Theorems

Q008 MCQ

Expand $(a + b)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$.

Correct option: C**Final answer**

$$a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

Q009 Written

Expand $(a - 2b)^4$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a - 2b)^4 = a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$.

Final answer

$$a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$$

Worked Solutions

Binomial and Multinomial Theorems

Q010 MCQ

Expand $(x - 2)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 2)^5 = x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$.

Correct option: A**Final answer**

$$x^5 - 10x^4 + 40x^3 - 80x^2 + 80x - 32$$

Q011 Written

Expand $(3x + 2y)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3x + 2y)^5 = 243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$.

Final answer

$$243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q012 Written

Expand $(2a^2 - 1)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a^2 - 1)^5 = 32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$.

Final answer

$$32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$$

Q013 Written

Expand $(x - 3)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 3)^6 = x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$.

Final answer

$$x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$$

Worked Solutions

Binomial and Multinomial Theorems

Q014 Written

Expand $(2x - 3y)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x - 3y)^6 = 64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$.

Final answer

$$64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$$

Q015 Written

Expand $\left(x + \frac{1}{3}\right)^6$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $\left(x + \frac{1}{3}\right)^6 = x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$.

Final answer

$$x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$$

Worked Solutions

Binomial and Multinomial Theorems

Q016 MCQ

Expand $(x + 3)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 3)^3 = x^3 + 9x^2 + 27x + 27$.

Correct option: B**Final answer**

$$x^3 + 9x^2 + 27x + 27$$

Worked Solutions

Binomial and Multinomial Theorems

Q017 MCQ

Expand $(a + 2b)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a + 2b)^5 = a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$.

Correct option: D**Final answer**

$$a^5 + 10a^4b + 40a^3b^2 + 80a^2b^3 + 80ab^4 + 32b^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q018 MCQ

Expand $(x + 2)^2$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x + 2)^2 = x^2 + 4x + 4$.

Correct option: D**Final answer**

$$x^2 + 4x + 4$$

Q019 MCQ

Expand $(a - b)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$.

Correct option: A**Final answer**

$$a^3 - 3a^2b + 3ab^2 - b^3$$

Worked Solutions

Binomial and Multinomial Theorems

Q020 MCQ

Expand $(2x + 1)^5$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x + 1)^5 = 32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$.

Correct option: B**Final answer**

$$32x^5 + 80x^4 + 80x^3 + 40x^2 + 10x + 1$$

Q021 Written

Expand $(3a - 2b)^5$ using the Binomial Theorem.**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3a - 2b)^5 = 243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$.

Final answer

$$243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$$

Worked Solutions

Binomial and Multinomial Theorems

Q022 MCQ

Expand $(2x - 1)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2x - 1)^4 = 16x^4 - 32x^3 + 24x^2 - 8x + 1$.

Correct option: C**Final answer**

$$16x^4 - 32x^3 + 24x^2 - 8x + 1$$

Q023 MCQ

Expand $(x - 3)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(x - 3)^4 = x^4 - 12x^3 + 54x^2 - 108x + 81$.

Correct option: B**Final answer**

$$x^4 - 12x^3 + 54x^2 - 108x + 81$$

Worked Solutions

Binomial and Multinomial Theorems

Q024 MCQ

Expand $(2a + b)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a + b)^3 = 8a^3 + 12a^2b + 6ab^2 + b^3$.

Correct option: C**Final answer**

$$8a^3 + 12a^2b + 6ab^2 + b^3$$

Q025 MCQ

Expand $(2a^2 + 3)^4$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(2a^2 + 3)^4 = 16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$.

Correct option: C**Final answer**

$$16a^8 + 96a^6 + 216a^4 + 216a^2 + 81$$

Worked Solutions

Binomial and Multinomial Theorems

Q026 MCQ

Expand $(3a^2 - 2)^3$ using the Binomial Theorem:**Solution**

1. Write the terms from the first factor power n down to 0.
2. After simplifying, $(3a^2 - 2)^3 = 27a^6 - 54a^4 + 36a^2 - 8$.

Correct option: B**Final answer**

$$27a^6 - 54a^4 + 36a^2 - 8$$

Q027 Written

Find the coefficient of a^3b^2 in the expansion of $(a + b)^5$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C_2 times the first factor to power q and the second factor to power r .
3. The required coefficient is 10.

Final answer

10

Worked Solutions

Binomial and Multinomial Theorems

Q028 Written

Find the coefficient of x^4y^2 in the expansion of $(x - 2y)^6$.**Solution**

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 2$.
2. Use $6C2$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 60.

Final answer

60

Q029 Written

Find the coefficient of x^4 in the expansion of $(2x^2 - 3)^7$.**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 5$.
2. Use $7C5$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -20412 .

Final answer -20412

Worked Solutions

Binomial and Multinomial Theorems

Q030 Written

Find the coefficient of x^6y in the expansion of $(2x - 3y)^7$.**Solution**

1. The target power uses $q = 6$ copies of the first term, so $r = n - q = 1$.
2. Use 7C_1 times the first factor to power q and the second factor to power r .
3. The required coefficient is -1344 .

Final answer -1344

Q031 MCQ

Find the coefficient of x^2 in the expansion of $(3x - 2)^5$:**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 3$.
2. Use 5C_3 times the first factor to power q and the second factor to power r .
3. The required coefficient is -720 .

Correct option: B**Final answer** -720

Worked Solutions

Binomial and Multinomial Theorems

Q032 Written

Find the coefficient of a^6 in the expansion of $(3a^2 - 2)^8$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 5$.
2. Use $8C5$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -48384 .

Final answer -48384

Q033 MCQ

Find the coefficient of x^4y^4 in the expansion of $(2x - y)^8$:**Solution**

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 4$.
2. Use $8C4$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 1120.

Correct option: A**Final answer**

1120

Worked Solutions

Binomial and Multinomial Theorems

Q034 MCQ

Find the coefficient of ab^3 in the expansion of $(3a + b)^4$:**Solution**

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use $4C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 12.

Correct option: B**Final answer**

12

Q035 MCQ

Find the coefficient of x^3y^2 in the expansion of $(x - 4y)^5$:**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use $5C2$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 160.

Correct option: C**Final answer**

160

Worked Solutions

Binomial and Multinomial Theorems

Q036 MCQ

Find the coefficient of x^4 in the expansion of $(3x + 1)^5$:**Solution**

1. The target power uses $q = 4$ copies of the first term, so $r = n - q = 1$.
2. Use $5C1$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 405.

Correct option: D**Final answer**

405

Q037 Written

Find the coefficient of a in the expansion of $(2a - 3)^4$.**Solution**

1. The target power uses $q = 1$ copies of the first term, so $r = n - q = 3$.
2. Use $4C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -216 .

Final answer -216

Worked Solutions

Binomial and Multinomial Theorems

Q038 Written

Find the coefficient of x^3 in the expansion of $(3x + 2)^5$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 2$.
2. Use 5C_2 times the first factor to power q and the second factor to power r .
3. The required coefficient is 1080.

Final answer

1080

Q039 Written

Find the coefficient of x^6 in the expansion of $(2x^2 - 1)^6$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use 6C_3 times the first factor to power q and the second factor to power r .
3. The required coefficient is -160 .

Final answer -160

Worked Solutions

Binomial and Multinomial Theorems

Q040 MCQ

Find the coefficient of x^4 in the expansion of $(3x^2 - 1)^5$:**Solution**

1. The target power uses $q = 2$ copies of the first term, so $r = n - q = 3$.
2. Use $5C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is -90 .

Correct option: D**Final answer** -90

Q041 Written

Find the coefficient of x^6 in the expansion of $(3x^2 + 2)^6$.**Solution**

1. The target power uses $q = 3$ copies of the first term, so $r = n - q = 3$.
2. Use $6C3$ times the first factor to power q and the second factor to power r .
3. The required coefficient is 4320 .

Final answer 4320

Worked Solutions

Binomial and Multinomial Theorems

Q042 Written

If $(x + 1)^5 = a_0 + a_x + \dots$, find the sum of its coefficients.

Solution

1. Put $x = 1$; the sum of the coefficients is the value of the expansion at $x = 1$.
2. Therefore the required value is 32.

Final answer

32

Q043 MCQ

Find the coefficient of xyz in the expansion of $(x + y + z)^3$.

Solution

1. The target powers are $p = 1$, $q = 1$, $r = 1$, and $p + q + r = 3$.
2. Coefficient = $\frac{3!}{1! \times 1! \times 1!} \times (1)^1 \times (1)^1 \times (1)^1 = 6$.
3. Therefore, the required coefficient is 6.

Correct option: A

Final answer

6

Worked Solutions

Binomial and Multinomial Theorems

Q044 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a + b + c)^8$:**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is 280.

Correct option: C**Final answer**

280

Q045 MCQ

Find the coefficient of a^3bc^2 in the expansion of $(a + b + c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is 60.

Correct option: A**Final answer**

60

Worked Solutions

Binomial and Multinomial Theorems

Q046 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + b + c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 90.

Correct option: B**Final answer**

90

Q047 MCQ

Find the coefficient of xy^5z in the expansion of $(x + y + z)^7$:**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is 42.

Correct option: A**Final answer**

42

Worked Solutions

Binomial and Multinomial Theorems

Q048 Written

Find the coefficient of x^4y^2z in the expansion of $(x + y + z)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is 105.

Final answer

105

Q049 Written

Find the coefficient of $x^2y^3z^2$ in the expansion of $(x + y + z)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 210.

Final answer

210

Worked Solutions

Binomial and Multinomial Theorems

Q050 Written

Find the coefficient of x^6yz^2 in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 252$ on the signed factors.
3. The required coefficient is 252.

Final answer

252

Q051 Written

Find the coefficient of $x^3y^3z^3$ in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 1680$ on the signed factors.
3. The required coefficient is 1680.

Final answer

1680

Worked Solutions

Binomial and Multinomial Theorems

Q052 Written

Find the coefficient of $x^2y^4z^3$ in the expansion of $(x + y + z)^9$.**Solution**

1. The target exponents add to 9; compare this with $n = 9$.
2. Use $\frac{n!}{p! \times q! \times r!} = 1260$ on the signed factors.
3. The required coefficient is 1260.

Final answer

1260

Q053 MCQ

Find the coefficient of a^4b^3c in the expansion of $(a - 2b + 3c)^8$:**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -6720 .

Correct option: D**Final answer** -6720

Worked Solutions

Binomial and Multinomial Theorems

Q054 MCQ

Find the coefficient of $a^2b^2c^2$ in the expansion of $(a + 3b - 2c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 90$ on the signed factors.
3. The required coefficient is 3240.

Correct option: C**Final answer**

3240

Q055 MCQ

Find the coefficient of ab^2c^3 in the expansion of $(a + 3b - 2c)^6$:**Solution**

1. The target exponents add to 6; compare this with $n = 6$.
2. Use $\frac{n!}{p! \times q! \times r!} = 60$ on the signed factors.
3. The required coefficient is -4320 .

Correct option: D**Final answer** -4320

Worked Solutions

Binomial and Multinomial Theorems

Q056 Written

Find the coefficient of $a^2b^3c^2$ in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 210$ on the signed factors.
3. The required coefficient is 15120.

Final answer

15120

Q057 Written

Find the coefficient of abc^5 in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 42$ on the signed factors.
3. The required coefficient is -20412 .

Final answer -20412

Worked Solutions

Binomial and Multinomial Theorems

Q058 Written

Find the coefficient of a^2b^4c in the expansion of $(a + 2b - 3c)^7$.**Solution**

1. The target exponents add to 7; compare this with $n = 7$.
2. Use $\frac{n!}{p! \times q! \times r!} = 105$ on the signed factors.
3. The required coefficient is -5040 .

Final answer -5040

Q059 Written

Find the coefficient of $a^2b^2c^4$ in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 420$ on the signed factors.
3. The required coefficient is 136080.

Final answer

136080

Worked Solutions

Binomial and Multinomial Theorems

Q060 Written

Find the coefficient of a^4bc^3 in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 280$ on the signed factors.
3. The required coefficient is -120960 .

Final answer -120960

Q061 Written

Find the coefficient of $a^3b^3c^2$ in the expansion of $(2a - b + 3c)^8$.**Solution**

1. The target exponents add to 8; compare this with $n = 8$.
2. Use $\frac{n!}{p! \times q! \times r!} = 560$ on the signed factors.
3. The required coefficient is -40320 .

Final answer -40320

Worked Solutions

Binomial and Multinomial Theorems

Q062 Written

Find the greatest coefficient in the expansion of $(x^2 + x + 1)^4$.

Solution

1. Expand and collect like powers: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$.
2. The greatest coefficient is 19.

Final answer

19

Quiz MCQ 1 Concept Quiz · MCQ

The coefficient of x^4 in $(2x - 3)^5$ is:

Solution

1. $q = 4$ and $r = 1$, so the coefficient is -240 .

Correct option: A

Final answer -240

Worked Solutions

Binomial and Multinomial Theorems

Quiz MCQ 2 [Concept Quiz](#) · [MCQ](#)

The coefficient of $x^2y^2z^2$ in $(x + y + z)^6$ is:

Solution

1. Use $6!/(2!2!2!)$.

Correct option: C

Final answer

90

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)

The coefficient of x^6 in $(3x^2 - 2)^6$ is:

Solution

1. Choose the first factor three times.

Correct option: A

Final answer

−4320

Worked Solutions

Binomial and Multinomial Theorems

Quiz MCQ 4 Concept Quiz · MCQ

The coefficient of x^5 in $(2x^2 + 1)^4$ is:

Solution

1. All powers of x are even, so the coefficient is zero.

Correct option: A

Final answer

0

Quiz WRITTEN 5 Concept Quiz · Written

Expand $(2x - 1)^4$ fully.

Solution

1. The expansion is $16x^4 - 32x^3 + 24x^2 - 8x + 1$.

Final answer

$16x^4 - 32x^3 + 24x^2 - 8x + 1$

Worked Solutions

Binomial and Multinomial Theorems

Quiz WRITTEN 6 Concept Quiz · Written

Find the coefficient of x^4 in $(3x^2 + 2)^5$. Show the exponent check.

Solution

1. The first-term exponent is $q = 2$ and $r = 3$, giving ${}^5C_3 \times 9 \times 8 = 720$.

Final answer

720

Quiz WRITTEN 7 Concept Quiz · Written

Find the coefficient of x^2y^3z in $(x - 2y + z)^6$.

Solution

1. Use the multinomial coefficient and the signed factor; the result is -480 .

Final answer

 -480

Worked Solutions

Binomial and Multinomial Theorems

Quiz WRITTEN 8 Concept Quiz · Written

Find the greatest coefficient in $(x^2 + x + 1)^4$. Show enough work to justify the maximum.

Solution

1. Expand: $x^8 + 4x^7 + 10x^6 + 16x^5 + 19x^4 + 16x^3 + 10x^2 + 4x + 1$; the greatest coefficient is 19.

Final answer

19

Answer key

Use the question number shown on each page.

Q001 A	Q018 D	Q036 D	Q054 C
Q002 A	Q019 A	Q037 -216	Q055 D
Q003 A	Q020 B	Q038 1080	Q056 15120
Q004 $x^5 + 10x^4 + 40x^3 + 80x^2 + 80x + 32$	Q021 $243a^5 - 810a^4b + 1080a^3b^2 - 720a^2b^3 + 240ab^4 - 32b^5$	Q039 -160	Q057 -20412
Q005 D	Q022 C	Q040 D	Q058 -5040
Q006 $x^6 + 12x^5 + 60x^4 + 160x^3 + 240x^2 + 192x + 64$	Q023 B	Q041 4320	Q059 136080
Q007 B	Q024 C	Q042 32	Q060 -120960
Q008 C	Q025 C	Q043 A	Q061 -40320
Q009 $a^4 - 8a^3b + 24a^2b^2 - 32ab^3 + 16b^4$	Q026 B	Q044 C	Q062 19
Q010 A	Q027 10	Q045 A	Quiz MCQ 1 A
Q011 $243x^5 + 810x^4y + 1080x^3y^2 + 720x^2y^3 + 240xy^4 + 32y^5$	Q028 60	Q046 B	Quiz MCQ 2 C
Q012 $32a^{10} - 80a^8 + 80a^6 - 40a^4 + 10a^2 - 1$	Q029 -20412	Q047 A	Quiz MCQ 3 A
Q013 $x^6 - 18x^5 + 135x^4 - 540x^3 + 1215x^2 - 1458x + 729$	Q030 -1344	Q048 105	Quiz MCQ 4 A
Q014 $64x^6 - 576x^5y + 2160x^4y^2 - 4320x^3y^3 + 4860x^2y^4 - 2916xy^5 + 729y^6$	Q031 B	Q049 210	Quiz WRITTEN 5 $16x^4 - 32x^3 + 24x^2 - 8x + 1$
Q015 $x^6 + 2x^5 + \frac{5x^4}{3} + \frac{20x^3}{27} + \frac{5x^2}{27} + \frac{2x}{81} + \frac{1}{729}$	Q032 -48384	Q050 252	Quiz WRITTEN 6 720
Q016 B	Q033 A	Q051 1680	Quiz WRITTEN 7 -480
Q017 D	Q034 B	Q052 1260	Quiz WRITTEN 8 19
	Q035 C	Q053 D	

Concept 2 | Polynomial Division

English Student Edition • Focused formulas and guided practice

Formula opener

Division identity

$$A(x) = B(x)Q(x) + R(x)$$

Remainder

$$R = P(a) \quad \text{for divisor } x - a$$

Degree check

$$\deg R < \deg B$$

Worked Solutions

Polynomial Division

Q001 Challenge

Find the quotient and the remainder when $2x^3 + x^2 - 7x + 3$ is divided by $x^2 + x - 2$. Then verify the result using $A = B \times Q + R$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 7x + 3 = (x^2 + x - 2) \times (2x - 1) + (-2x + 1)$.
4. Therefore, quotient $Q = 2x - 1$ and remainder $R = -2x + 1$.

Correct option: B

Final answer

$$Q = 2x - 1, \quad R = -2x + 1$$

Worked Solutions

Polynomial Division

Q002 Challenge

Find the polynomial A when it is divided by $x^2 - 2x + 3$, the quotient is $3x - 1$, and the remainder is $x + 2$. Then divide your answer back to check it:

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x + 3) \times (3x - 1) + (x + 2)$.
3. Expand and collect like powers: $A = 3x^3 - 7x^2 + 12x - 1$.
4. Therefore, $A = 3x^3 - 7x^2 + 12x - 1$.

Correct option: A

Final answer

$$3x^3 - 7x^2 + 12x - 1$$

Worked Solutions

Polynomial Division

Q003 Challenge

When $x^3 + 5x^2 + 7x - 7$ is divided by B, the quotient is $x + 3$ and the remainder is $2x - 4$. Find B. Then verify the degree condition:

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3+5x^2+7x-7-(2x-4)}{x+3}$.
3. Divide and simplify: $B = x^2 + 2x - 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 + 5x^2 + 7x - 7$.

Correct option: D

Final answer

$$x^2 + 2x - 1$$

Worked Solutions

Polynomial Division

Q004 Challenge

A cuboid has volume $2x^3 + 9x^2 + 4x - 15$ and base area $x^2 + 2x - 3$. Write its height in terms of x . State the mathematical check Which of the following is correct?:

Solution

1. Use height = volume $\div$ base area: $h = \frac{2x^3+9x^2+4x-15}{x^2+2x-3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 5$.

Correct option: C

Final answer

$$2x + 5$$

Worked Solutions

Polynomial Division

Q005 Challenge

Find the quotient and the remainder when $3x^4 - x^3 + 2x^2 - 4x + 1$ is divided by $x^2 - 2x + 2$. Decide whether the result is exact and justify from R:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - x^3 + 2x^2 - 4x + 1 = (x^2 - 2x + 2) \times (3x^2 + 5x + 6) + (-2x - 11)$.
4. Therefore, quotient $Q = 3x^2 + 5x + 6$ and remainder $R = -2x - 11$.

Correct option: B

Final answer

$$Q = 3x^2 + 5x + 6, \quad R = -2x - 11$$

Worked Solutions

Polynomial Division

Q006 Challenge

Find the quotient and the remainder when $x^4 - 3x^2 + 2x + 6$ is divided by $x + 2$. Explain why a zero placeholder is needed:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^2 + 2x + 6 = (x + 2) \times (x^3 - 2x^2 + x) + (6)$.
4. Therefore, quotient $Q = x^3 - 2x^2 + x$ and remainder $R = 6$.

Correct option: A

Final answer

$$Q = x^3 - 2x^2 + x, \quad R = 6$$

Worked Solutions

Polynomial Division

Q007 Written

Find the quotient and the remainder when $x^2 - 5x + 1$ is divided by $x - 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 - 5x + 1 = (x - 2) \times (x - 3) + (-5)$.
4. Therefore, quotient $Q = x - 3$ and remainder $R = -5$.

Final answer

$$Q = x - 3, \quad R = -5$$

Q008 Written

Find the quotient and the remainder when $x^3 - 4x + 1$ is divided by $x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 4x + 1 = (x - 3) \times (x^2 + 3x + 5) + (16)$.
4. Therefore, quotient $Q = x^2 + 3x + 5$ and remainder $R = 16$.

Final answer

$$Q = x^2 + 3x + 5, \quad R = 16$$

Worked Solutions

Polynomial Division

Q009 MCQ

Find the quotient and the remainder when $3x^3 + 2x^2 - 7x + 5$ is divided by $x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 2x^2 - 7x + 5 = (x - 2) \times (3x^2 + 8x + 9) + (23)$.
4. Therefore, quotient $Q = 3x^2 + 8x + 9$ and remainder $R = 23$.

Correct option: D**Final answer**

$$Q = 3x^2 + 8x + 9, \quad R = 23$$

Worked Solutions

Polynomial Division

Q010 MCQ

Find the quotient and the remainder when $x^3 + 6x^2 - 4$ is divided by $2x - 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 6x^2 - 4 = (2x - 3) \times \left(\frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8} \right) + \left(\frac{103}{8} \right)$.
4. Therefore, quotient $Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}$ and remainder $R = \frac{103}{8}$.

Correct option: B**Final answer**

$$Q = \frac{x^2}{2} + \frac{15x}{4} + \frac{45}{8}, \quad R = \frac{103}{8}$$

Worked Solutions

Polynomial Division

Q011 MCQ

Find the quotient and the remainder when $3x^3 - 4x^2 + 2x + 7$ is divided by $x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 4x^2 + 2x + 7 = (x - 1) \times (3x^2 - x + 1) + (8)$.
4. Therefore, quotient $Q = 3x^2 - x + 1$ and remainder $R = 8$.

Correct option: B**Final answer**

$$Q = 3x^2 - x + 1, \quad R = 8$$

Q012 MCQ

Find the quotient and the remainder when $2x^4 + x^3 - 5x + 4$ is divided by $x + 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 + x^3 - 5x + 4 = (x + 2) \times (2x^3 - 3x^2 + 6x - 17) + (38)$.
4. Therefore, quotient $Q = 2x^3 - 3x^2 + 6x - 17$ and remainder $R = 38$.

Correct option: A**Final answer**

$$Q = 2x^3 - 3x^2 + 6x - 17, \quad R = 38$$

Worked Solutions

Polynomial Division

Q013 MCQ

Find the quotient and the remainder when $4x^4 - 3x^3 + x - 6$ is divided by $x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^4 - 3x^3 + x - 6 = (x - 2) \times (4x^3 + 5x^2 + 10x + 21) + (36)$.
4. Therefore, quotient $Q = 4x^3 + 5x^2 + 10x + 21$ and remainder $R = 36$.

Correct option: D**Final answer**

$$Q = 4x^3 + 5x^2 + 10x + 21, \quad R = 36$$

Worked Solutions

Polynomial Division

Q014 MCQ

Find the quotient and the remainder when $2x^3 - 9x + 1$ is divided by $3x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 9x + 1 = (3x - 2) \times \left(\frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27} \right) + \left(-\frac{119}{27} \right)$.
4. Therefore, quotient $Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}$ and remainder $R = -\frac{119}{27}$.

Correct option: B**Final answer**

$$Q = \frac{2x^2}{3} + \frac{4x}{9} - \frac{73}{27}, \quad R = -\frac{119}{27}$$

Worked Solutions

Polynomial Division

Q015 Written

Find the quotient and the remainder when $2x^3 - 5x^2 - 2$ is divided by $2x^2 - x - 1$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 5x^2 - 2 = (2x^2 - x - 1) \times (x - 2) + (-x - 4)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = -x - 4$.

Final answer

$$Q = x - 2, \quad R = -x - 4$$

Q016 Written

Find the quotient and the remainder when $2x^3 - 13x + 1$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 13x + 1 = (x^2 + 2x - 3) \times (2x - 4) + (x - 11)$.
4. Therefore, quotient $Q = 2x - 4$ and remainder $R = x - 11$.

Final answer

$$Q = 2x - 4, \quad R = x - 11$$

Worked Solutions

Polynomial Division

Q017 Written

Find the quotient and the remainder when $2x^3 - 3x^2 + 3x + 4$ is divided by $x^2 - 2x + 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 - 3x^2 + 3x + 4 = (x^2 - 2x + 3) \times (2x + 1) + (-x + 1)$.
4. Therefore, quotient $Q = 2x + 1$ and remainder $R = -x + 1$.

Final answer

$$Q = 2x + 1, \quad R = -x + 1$$

Q018 Written

Find the quotient and the remainder when $x^3 - 7x + 6$ is divided by $x^2 + 2x - 3$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 7x + 6 = (x^2 + 2x - 3) \times (x - 2) + (0)$.
4. Therefore, quotient $Q = x - 2$ and remainder $R = 0$.

Final answer

$$Q = x - 2, \quad R = 0$$

Worked Solutions

Polynomial Division

Q019 MCQ

Find the quotient and the remainder when $x^3 + 2x^2 - 5x + 4$ is divided by $x^2 + x - 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 2x^2 - 5x + 4 = (x^2 + x - 2) \times (x + 1) + (-4x + 6)$.
4. Therefore, quotient $Q = x + 1$ and remainder $R = -4x + 6$.

Correct option: D**Final answer**

$$Q = x + 1, \quad R = -4x + 6$$

Q020 MCQ

Find the quotient and the remainder when $x^4 - 3x^3 + 2x - 5$ is divided by $x^2 + 2x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 3x^3 + 2x - 5 = (x^2 + 2x - 1) \times (x^2 - 5x + 11) + (-25x + 6)$.
4. Therefore, quotient $Q = x^2 - 5x + 11$ and remainder $R = -25x + 6$.

Correct option: B**Final answer**

$$Q = x^2 - 5x + 11, \quad R = -25x + 6$$

Worked Solutions

Polynomial Division

Q021 MCQ

Find the quotient and the remainder when $2x^3 + x^2 - 4x + 5$ is divided by $x^2 - 2x + 2$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^3 + x^2 - 4x + 5 = (x^2 - 2x + 2) \times (2x + 5) + (2x - 5)$.
4. Therefore, quotient $Q = 2x + 5$ and remainder $R = 2x - 5$.

Correct option: D**Final answer**

$$Q = 2x + 5, \quad R = 2x - 5$$

Q022 MCQ

Find the quotient and the remainder when $x^4 - 2x^3 + 5x^2 - x + 3$ is divided by $x^2 + 3x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^4 - 2x^3 + 5x^2 - x + 3 = (x^2 + 3x - 1) \times (x^2 - 5x + 21) + (-69x + 24)$.
4. Therefore, quotient $Q = x^2 - 5x + 21$ and remainder $R = -69x + 24$.

Correct option: C**Final answer**

$$Q = x^2 - 5x + 21, \quad R = -69x + 24$$

Worked Solutions

Polynomial Division

Q023 MCQ

Find the quotient and the remainder when $3x^3 - 5x + 2$ is divided by $x^2 - x - 1$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 - 5x + 2 = (x^2 - x - 1) \times (3x + 3) + (x + 5)$.
4. Therefore, quotient $Q = 3x + 3$ and remainder $R = x + 5$.

Correct option: B**Final answer**

$$Q = 3x + 3, \quad R = x + 5$$

Q024 MCQ

Find the quotient and the remainder when $2x^4 - x^3 + 4x^2 - 7$ is divided by $x^2 + x + 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $2x^4 - x^3 + 4x^2 - 7 = (x^2 + x + 3) \times (2x^2 - 3x + 1) + (8x - 10)$.
4. Therefore, quotient $Q = 2x^2 - 3x + 1$ and remainder $R = 8x - 10$.

Correct option: A**Final answer**

$$Q = 2x^2 - 3x + 1, \quad R = 8x - 10$$

Worked Solutions

Polynomial Division

Q025 Written

Find the quotient and the remainder when $6x^3 - 8x^2 + 5$ is divided by $2x^2 - 2x - 5$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 8x^2 + 5 = (2x^2 - 2x - 5) \times (3x - 1) + (13x)$.
4. Therefore, quotient $Q = 3x - 1$ and remainder $R = 13x$.

Final answer

$$Q = 3x - 1, \quad R = 13x$$

Q026 Written

Find the quotient and the remainder when $3x^4 - 5x^3 + x - 1$ is divided by $-x^2 + 3x + 2$.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 - 5x^3 + x - 1 = (-x^2 + 3x + 2) \times (-3x^2 - 4x - 18) + (63x + 35)$.
4. Therefore, quotient $Q = -3x^2 - 4x - 18$ and remainder $R = 63x + 35$.

Final answer

$$Q = -3x^2 - 4x - 18, \quad R = 63x + 35$$

Worked Solutions

Polynomial Division

Q027 MCQ

Find the quotient and the remainder when $4x^3 - x^2 + 8x - 3$ is divided by $2x^2 + x - 4$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $4x^3 - x^2 + 8x - 3 = (2x^2 + x - 4) \times \left(2x - \frac{3}{2}\right) + \left(\frac{35x}{2} - 9\right)$.
4. Therefore, quotient $Q = 2x - \frac{3}{2}$ and remainder $R = \frac{35x}{2} - 9$.

Correct option: D**Final answer**

$$Q = 2x - \frac{3}{2}, \quad R = \frac{35x}{2} - 9$$

Q028 MCQ

Find the quotient and the remainder when $x^3 - 8x + 2$ is divided by $x^2 - 4$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 - 8x + 2 = (x^2 - 4) \times (x) + (-4x + 2)$.
4. Therefore, quotient $Q = x$ and remainder $R = -4x + 2$.

Correct option: B**Final answer**

$$Q = x, \quad R = -4x + 2$$

Worked Solutions

Polynomial Division

Q029 MCQ

Find the quotient and the remainder when $5x^3 - 4x^2 + x - 6$ is divided by $2x^2 - x + 1$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $5x^3 - 4x^2 + x - 6 = (2x^2 - x + 1) \times \left(\frac{5x}{2} - \frac{3}{4}\right) + \left(-\frac{9x}{4} - \frac{21}{4}\right)$.
4. Therefore, quotient $Q = \frac{5x}{2} - \frac{3}{4}$ and remainder $R = -\frac{9x}{4} - \frac{21}{4}$.

Correct option: D

Final answer

$$Q = \frac{5x}{2} - \frac{3}{4}, \quad R = -\frac{9x}{4} - \frac{21}{4}$$

Worked Solutions

Polynomial Division

Q030 MCQ

Find the quotient and the remainder when $3x^4 + 2x^3 - x^2 + 5$ is divided by $4x^2 + x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^4 + 2x^3 - x^2 + 5 = (4x^2 + x - 2) \times \left(\frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64} \right) + \left(\frac{37x}{64} + \frac{163}{32} \right)$.
4. Therefore, quotient $Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}$ and remainder $R = \frac{37x}{64} + \frac{163}{32}$.

Correct option: C

Final answer

$$Q = \frac{3x^2}{4} + \frac{5x}{16} + \frac{3}{64}, \quad R = \frac{37x}{64} + \frac{163}{32}$$

Worked Solutions

Polynomial Division

Q031 MCQ

Find the quotient and the remainder when $3x^3 + 4x^2 - x + 9$ is divided by $2x^2 - x + 3$:**Solution**

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3x^3 + 4x^2 - x + 9 = (2x^2 - x + 3) \times \left(\frac{3x}{2} + \frac{11}{4}\right) + \left(-\frac{11x}{4} + \frac{3}{4}\right)$.
4. Therefore, quotient $Q = \frac{3x}{2} + \frac{11}{4}$ and remainder $R = -\frac{11x}{4} + \frac{3}{4}$.

Correct option: D**Final answer**

$$Q = \frac{3x}{2} + \frac{11}{4}, \quad R = -\frac{11x}{4} + \frac{3}{4}$$

Worked Solutions

Polynomial Division

Q032 MCQ

Find the quotient and the remainder when $6x^3 - 5x + 1$ is divided by $2x^2 + 3x - 2$:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $6x^3 - 5x + 1 = (2x^2 + 3x - 2) \times \left(3x - \frac{9}{2}\right) + \left(\frac{29x}{2} - 8\right)$.
4. Therefore, quotient $Q = 3x - \frac{9}{2}$ and remainder $R = \frac{29x}{2} - 8$.

Correct option: B

Final answer

$$Q = 3x - \frac{9}{2}, \quad R = \frac{29x}{2} - 8$$

Q033 Written

Find the quotient and the remainder when $-9a^2 - 9ax + 4x^2$ is divided by $-3a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-9a^2 - 9ax + 4x^2 = (-3a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions

Polynomial Division

Q034 Written

Find the quotient and the remainder when $-5a^3 + 5a^2x - 3ax^2 + x^3$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-5a^3 + 5a^2x - 3ax^2 + x^3 = (-2a + x) \times (3a^2 - ax + x^2) + (a^3)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = a^3$.

Final answer

$$Q = 3a^2 - ax + x^2, \quad R = a^3$$

Q035 Written

Find the quotient and the remainder when $-6a^2 - 5ax + 4x^2$ is divided by $-2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-6a^2 - 5ax + 4x^2 = (-2a + x) \times (3a + 4x) + (0)$.
4. Therefore, quotient $Q = 3a + 4x$ and remainder $R = 0$.

Final answer

$$Q = 3a + 4x, \quad R = 0$$

Worked Solutions

Polynomial Division

Q036 Written

Find the quotient and the remainder when $-a^3 + a^2x + ax^2 + x^3$ is divided by $2a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-a^3 + a^2x + ax^2 + x^3 = (2a + x) \times (3a^2 - ax + x^2) + (-7a^3)$.
4. Therefore, quotient $Q = 3a^2 - ax + x^2$ and remainder $R = -7a^3$.

Final answer

$$Q = 3a^2 - ax + x^2, \quad R = -7a^3$$

Q037 Written

Find the quotient and the remainder when $3a^3 - 2ax^2 + x^3$ is divided by $a + x$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $3a^3 - 2ax^2 + x^3 = (a + x) \times (3a^2 - 3ax + x^2) + (0)$.
4. Therefore, quotient $Q = 3a^2 - 3ax + x^2$ and remainder $R = 0$.

Final answer

$$Q = 3a^2 - 3ax + x^2, \quad R = 0$$

Worked Solutions

Polynomial Division

Q038 MCQ

Find the quotient and the remainder when $-2a^3 + 7a^2x - 4ax^2 + x^3$ is divided by $-a + x$. Treat it as a polynomial in x ; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $-2a^3 + 7a^2x - 4ax^2 + x^3 = (-a + x) \times (4a^2 - 3ax + x^2) + (2a^3)$.
4. Therefore, quotient $Q = 4a^2 - 3ax + x^2$ and remainder $R = 2a^3$.

Correct option: D

Final answer

$$Q = 4a^2 - 3ax + x^2, \quad R = 2a^3$$

Q039 MCQ

Find the quotient and the remainder when $x^2 + 3xy - 4y^2$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants:

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^2 + 3xy - 4y^2 = (x + y) \times (x + 2y) + (-6y^2)$.
4. Therefore, quotient $Q = x + 2y$ and remainder $R = -6y^2$.

Correct option: B

Final answer

$$Q = x + 2y, \quad R = -6y^2$$

Worked Solutions

Polynomial Division

Q040 Written

Find the quotient and the remainder when $x^3 + 3x^2y - 2y^3$ is divided by $x + y$. Treat it as a polynomial in x ; other letters are constants.

Solution

1. Arrange the dividend and divisor in descending powers of x ; insert a zero coefficient for any missing power. Treat it as a polynomial in x ; other letters are constants.
2. Use polynomial long division until the remaining degree is lower than the divisor degree.
3. The identity check is $x^3 + 3x^2y - 2y^3 = (x + y) \times (x^2 + 2xy - 2y^2) + (0)$.
4. Therefore, quotient $Q = x^2 + 2xy - 2y^2$ and remainder $R = 0$.

Final answer

$$Q = x^2 + 2xy - 2y^2, \quad R = 0$$

Q041 Written

Find the polynomial A when it is divided by $x^2 - 2x - 1$, the quotient is $x + 3$, and the remainder is $-2x + 5$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x - 1) \times (x + 3) + (-2x + 5)$.
3. Expand and collect like powers: $A = x^3 + x^2 - 9x + 2$.
4. Therefore, $A = x^3 + x^2 - 9x + 2$.

Final answer

$$x^3 + x^2 - 9x + 2$$

Worked Solutions

Polynomial Division

Q042 Written

Find the polynomial **A** when it is divided by $x^2 - 2x - 1$, the quotient is $2x - 3$, and the remainder is $-2x$.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 2x - 1) \times (2x - 3) + (-2x)$.
3. Expand and collect like powers: $A = 2x^3 - 7x^2 + 2x + 3$.
4. Therefore, $A = 2x^3 - 7x^2 + 2x + 3$.

Final answer

$$2x^3 - 7x^2 + 2x + 3$$

Q043 Written

Find the polynomial **A** when it is divided by $x^2 - 3x + 2$, the quotient is $2x + 6$, and the remainder is $2x - 3$.

Solution

1. Use the quotient–remainder identity $A = B \times Q + R$.
2. Substitute: $A = (x^2 - 3x + 2) \times (2x + 6) + (2x - 3)$.
3. Expand and collect like powers: $A = 2x^3 - 12x + 9$.
4. Therefore, $A = 2x^3 - 12x + 9$.

Final answer

$$2x^3 - 12x + 9$$

Worked Solutions

Polynomial Division

Q044 Written

Find the polynomial A when it is divided by $2x^2 - x + 2$, the quotient is $2x - 1$, and the remainder is 2.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (2x^2 - x + 2) \times (2x - 1) + (2)$.
3. Expand and collect like powers: $A = 4x^3 - 4x^2 + 5x$.
4. Therefore, $A = 4x^3 - 4x^2 + 5x$.

Final answer

$$4x^3 - 4x^2 + 5x$$

Q045 Written

Find the polynomial A when it is divided by $2x^2 + x + 1$, the quotient is $3x + 2$, and the remainder is $-2x + 6$.

Solution

1. Use the quotient-remainder identity $A = B \times Q + R$.
2. Substitute: $A = (2x^2 + x + 1) \times (3x + 2) + (-2x + 6)$.
3. Expand and collect like powers: $A = 6x^3 + 7x^2 + 3x + 8$.
4. Therefore, $A = 6x^3 + 7x^2 + 3x + 8$.

Final answer

$$6x^3 + 7x^2 + 3x + 8$$

Worked Solutions

Polynomial Division

Q046 Written

When $4x^3 + 8x - 9$ is divided by B, the quotient is $2x - 3$ and the remainder is $x + 15$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{4x^3+8x-9-(x+15)}{2x-3}$.
3. Divide and simplify: $B = 2x^2 + 3x + 8$.
4. Check by multiplying B by Q and adding R to recover $A = 4x^3 + 8x - 9$.

Final answer

$$2x^2 + 3x + 8$$

Q047 Written

When $x^3 - x^2 + 3x + 1$ is divided by B, the quotient is $x^2 - 2x + 5$ and the remainder is -4 . Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^3-x^2+3x+1-(-4)}{x^2-2x+5}$.
3. Divide and simplify: $B = x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^3 - x^2 + 3x + 1$.

Final answer

$$x + 1$$

Worked Solutions

Polynomial Division

Q048 Written

When $2x^3 + x^2 - 3x + 5$ is divided by B, the quotient is $x + 1$ and the remainder is $-5x + 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+x^2-3x+5-(-5x+2)}{x+1}$.
3. Divide and simplify: $B = 2x^2 - x + 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + x^2 - 3x + 5$.

Final answer

$$2x^2 - x + 3$$

Q049 Written

When $6x^3 + 11x^2 - 2$ is divided by B, the quotient is $2x^2 + 3x - 1$ and the remainder is -1 . Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{6x^3+11x^2-2-(-1)}{2x^2+3x-1}$.
3. Divide and simplify: $B = 3x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = 6x^3 + 11x^2 - 2$.

Final answer

$$3x + 1$$

Worked Solutions

Polynomial Division

Q050 Written

When $2x^3 + 4x^2 + 7$ is divided by B, the quotient is $x + 2$ and the remainder is $3x + 13$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{2x^3+4x^2+7-(3x+13)}{x+2}$.
3. Divide and simplify: $B = 2x^2 - 3$.
4. Check by multiplying B by Q and adding R to recover $A = 2x^3 + 4x^2 + 7$.

Final answer

$$2x^2 - 3$$

Q051 Written

When $x^4 + x^2 + 3x - 1$ is divided by B, the quotient is $x^2 + x + 1$ and the remainder is $3x - 2$. Find B.

Solution

1. Start with $A = B \times Q + R$ and rearrange: $B = \frac{A-R}{Q}$.
2. Substitute the known expressions: $B = \frac{x^4+x^2+3x-1-(3x-2)}{x^2+x+1}$.
3. Divide and simplify: $B = x^2 - x + 1$.
4. Check by multiplying B by Q and adding R to recover $A = x^4 + x^2 + 3x - 1$.

Final answer

$$x^2 - x + 1$$

Worked Solutions

Polynomial Division

Q052 Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Write its height in terms of x .

Solution

1. Use height = volume $\div$ base area: $h = \frac{4x^3 + 4x^2 - 9x - 9}{2x^2 - x - 3}$.
2. Divide the polynomials and check that the remainder is zero.
3. Therefore, $h = 2x + 3$.

Final answer

$$2x + 3$$

Quiz MCQ 1 Concept Quiz · MCQ

For $x^2 - 5x + 1$ divided by $x - 2$, what is R ?:

Solution

1. Long division gives $R = -5$.

Correct option: A

Final answer

$$-5$$

Worked Solutions

Polynomial Division

Quiz MCQ 2 **Concept Quiz · MCQ**

In a polynomial in x , a is treated as... Which of the following is correct?:

Solution

1. Only powers of x determine the division order.

Correct option: B

Final answer

constant

Quiz MCQ 3 **Concept Quiz · MCQ**

To recover B from $A = B \times Q + R$, use... Which of the following is correct?:

Solution

1. Isolate B after moving R to the other side.

Correct option: C

Final answer

$$\frac{A - R}{Q}$$

Worked Solutions

Polynomial Division

Quiz MCQ 4 Concept Quiz · MCQ

If B has degree 2, a valid remainder can have degree...:

Solution

1. The remainder degree must be below 2.

Correct option: C

Final answer

1

Quiz WRITTEN 5 Concept Quiz · Written

Divide $3x^3 - 5x^2 + 2x - 7$ by $x - 2$. State Q, R, and verify $A = B \times Q + R$.

Solution

1. Long division gives $Q = 3x^2 + x + 4, R = 1$.
2. Substitution verifies $3x^3 - 5x^2 + 2x - 7 = (x - 2) \times (3x^2 + x + 4) + (1)$.

Final answer

$$Q = 3x^2 + x + 4, \quad R = 1$$

Worked Solutions

Polynomial Division

Quiz WRITTEN 6 Concept Quiz · Written

Recover A when $B=x^2 - x - 2$, $Q=2x + 3$, $R=-x + 1$.

Solution

1. Use $A = B \times Q + R$.
2. $A = 2x^3 + x^2 - 8x - 5$.

Final answer

$$2x^3 + x^2 - 8x - 5$$

Quiz WRITTEN 7 Concept Quiz · Written

A cuboid has volume $4x^3 + 4x^2 - 9x - 9$ and base area $2x^2 - x - 3$. Find its height.

Solution

1. Divide volume by base area: $h = 2x + 3$.
2. The remainder is zero, so the height is polynomial.

Final answer

$$2x + 3$$

Worked Solutions

Polynomial Division

Quiz WRITTEN 8 Concept Quiz · Written

Divide $x^4 - 2x^3 + 3x - 1$ by $x^2 - x + 1$ and explain why the remainder is valid.

Solution

1. Long division gives $Q = x^2 - x - 2$ and $R = 2x + 1$.
2. The remainder degree is less than 2, so it is valid even though it is nonzero.

Final answer

$$Q = x^2 - x - 2, \quad R = 2x + 1$$

Answer key

Use the question number shown on each page.

Q001 B	Q016 $Q = 2x - 4$, $R = x - 11$	Q031 D	Q046 $2x^2 + 3x + 8$
Q002 A	Q017 $Q = 2x + 1$, $R = -x + 1$	Q032 B	Q047 $x + 1$
Q003 D	Q018 $Q = x - 2$, $R = 0$	Q033 $Q = 3a + 4x$, $R = 0$	Q048 $2x^2 - x + 3$
Q004 C	Q019 D	Q034 $Q = 3a^2 - ax + x^2$, $R = a^3$	Q049 $3x + 1$
Q005 B	Q020 B	Q035 $Q = 3a + 4x$, $R = 0$	Q050 $2x^2 - 3$
Q006 A	Q021 D	Q036 $Q = 3a^2 - ax + x^2$, $R = -7a^3$	Q051 $x^2 - x + 1$
Q007 $Q = x - 3$, $R = -5$	Q022 C	Q037 $Q = 3a^2 - 3ax + x^2$, $R = 0$	Q052 $2x + 3$
Q008 $Q = x^2 + 3x + 5$, $R = 16$	Q023 B	Q038 D	Quiz MCQ 1 A
Q009 D	Q024 A	Q039 B	Quiz MCQ 2 B
Q010 B	Q025 $Q = 3x - 1$, $R = 13x$	Q040 $Q = x^2 + 2xy - 2y^2$, $R = 0$	Quiz MCQ 3 C
Q011 B	Q026 $Q = -3x^2 - 4x - 18$, $R = 63x + 35$	Q041 $x^3 + x^2 - 9x + 2$	Quiz MCQ 4 C
Q012 A	Q027 D	Q042 $2x^3 - 7x^2 + 2x + 3$	Quiz WRITTEN 5 $Q = 3x^2 + x + 4$, $R = 1$
Q013 D	Q028 B	Q043 $2x^3 - 12x + 9$	Quiz WRITTEN 6 $2x^3 + x^2 - 8x - 5$
Q014 B	Q029 D	Q044 $4x^3 - 4x^2 + 5x$	Quiz WRITTEN 7 $2x + 3$
Q015 $Q = x - 2$, $R = -x - 4$	Q030 C	Q045 $6x^3 + 7x^2 + 3x + 8$	Quiz WRITTEN 8 $Q = x^2 - x - 2$, $R = 2x + 1$

Concept 3 | Rational Expressions

English Student Edition • Focused formulas and guided practice

Formula opener

Restriction

denominator $\neq 0$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Worked Solutions

Rational Expressions

Q001 Challenge

After simplifying this transfer chain before substitution, which result is correct for 2():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2x(x+3)}{(x-3)(x-1)}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{2x(x+3)}{(x-3)(x-1)}$$

Worked Solutions

Rational Expressions

Q002 Challenge

After simplifying this transfer chain before substitution, which result is correct for 3():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3(x-3)(x+2)}{x(x+3)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{3(x-3)(x+2)}{x(x+3)}$$

Worked Solutions

Rational Expressions

Q003 Challenge

After simplifying this transfer chain before substitution, which result is correct for 4():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8(x-4)(x+1)}{(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{8(x-4)(x+1)}{(x-3)(x-2)}$$

Worked Solutions

Rational Expressions

Q004 Challenge

After simplifying this transfer chain before substitution, which result is correct for 8():?

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8}{x}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8}{x}$$

Worked Solutions

Rational Expressions

Q005 Challenge

After simplifying this transfer chain before substitution, which result is correct for 9():?

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$$

Worked Solutions

Rational Expressions

Q006 Challenge

After simplifying this transfer chain before substitution, which result is correct for 10():

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$$

Worked Solutions

Rational Expressions

Q007 Written

Simplify $\frac{12ab^5}{9a^3b^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{4b^3}{3a^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{4b^3}{3a^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{4b^3}{3a^2}$$

Q008 Written

Simplify $\frac{6x^2y^3}{20x^3y}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3y^2}{10x}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3y^2}{10x}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3y^2}{10x}$$

Worked Solutions

Rational Expressions

Q009 Written

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{3a}{4b^3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{3a}{4b^3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{3a}{4b^3}$$

Q010 Written

Simplify $\frac{2a^2b}{8a^3b^3}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{1}{4ab^2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{1}{4ab^2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{1}{4ab^2}$$

Worked Solutions

Rational Expressions

Q011 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4b^3}{a^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{4b^3}{a^2}$$

Q012 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9y^2}{10x}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9y^2}{10x}$$

Worked Solutions

Rational Expressions

Q013 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a}{b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a}{b^3}$$

Q014 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{1}{2ab^2}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{1}{2ab^2}$$

Worked Solutions

Rational Expressions

Q015 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3a^2}{2b^3}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{3a^2}{2b^3}$$

Q016 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{9x^2y^2}{4}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{9x^2y^2}{4}$$

Worked Solutions

Rational Expressions

Q017 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{3b^3}{a^3c}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{3b^3}{a^3c}$$

Q018 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{4x^3}{3y^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{4x^3}{3y^2}$$

Worked Solutions

Rational Expressions

Q019 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{2a^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: A**Final answer**

$$\frac{2a^3}{b^2}$$

Q020 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= 3x^2y^3$.
4. Retain the original denominator restrictions.

Correct option: B**Final answer**

$$3x^2y^3$$

Worked Solutions

Rational Expressions

Q021 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= -\frac{3c^2}{2a^3}$.
4. Retain the original denominator restrictions.

Correct option: C**Final answer**

$$-\frac{3c^2}{2a^3}$$

Q022 Written

Simplify $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

Solution

1. Factor each polynomial: $\frac{a^3b^2}{3c}$ and $\frac{9c^3}{a^4b^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{3c^2}{a}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{3c^2}{a}$$

Worked Solutions

Rational Expressions

Q023 Written

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

Solution

1. Factor each polynomial: $\frac{6bc^3}{5a^2}$ and $\frac{10a^4}{3b^3}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{4a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{4a^2c^3}{b^2}$$

Q024 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{12c^2}{a}$.
4. Retain the original denominator restrictions.

Correct option: C

Final answer

$$\frac{12c^2}{a}$$

Worked Solutions

Rational Expressions

Q025 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{8a^2c^3}{b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{8a^2c^3}{b^2}$$

Q026 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{27ac}{2b^2}$.
4. Retain the original denominator restrictions.

Correct option: D

Final answer

$$\frac{27ac}{2b^2}$$

Worked Solutions

Rational Expressions

Q027 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{9ax^2}{y}$.
4. Retain the original denominator restrictions.

Correct option: A

Final answer

$$\frac{9ax^2}{y}$$

Q028 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{64bc}{5a}$.
4. Retain the original denominator restrictions.

Correct option: B

Final answer

$$\frac{64bc}{5a}$$

Worked Solutions

Rational Expressions

Q029 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy}\right)$?:

Solution

1. Keep the same family first move; factor the expression.
2. Cancel complete factors only, never terms joined by addition.
3. Therefore the correct result is $= \frac{18abxy^2}{5}$.
4. Retain the original denominator restrictions.

Correct option: C**Final answer**

$$\frac{18abxy^2}{5}$$

Q030 Written

Simplify $\frac{x^2-4}{3x^2+7x+2}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x-2}{3x+1}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x-2}{3x+1}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions

Rational Expressions

Q031 Written

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2x-5}{x+2}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2x-5}{x+2}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2x-5}{x+2}$$

Q032 Written

Simplify $\frac{x^2+3x}{x^2-9}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x}{x-3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions

Rational Expressions

Q033 Written

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x+2}{x-4}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x+2}{x-4}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x+2}{x-4}$$

Q034 Written

Simplify $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.**Solution**

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{2(x-y)}{x(x+4y)}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{2(x-y)}{x(x+4y)}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{2(x-y)}{x(x+4y)}$$

Worked Solutions

Rational Expressions

Q035 Written

Simplify $\frac{x^3-1}{x^2+2x-3}$.

Solution

1. Factor the numerator and denominator before cancelling any factor.
2. The factored form is $\frac{x^2+x+1}{x+3}$; cancel only common factors.
3. Therefore the simplified expression is $= \frac{x^2+x+1}{x+3}$.
4. Keep every value excluded by an original denominator.

Final answer

$$\frac{x^2 + x + 1}{x + 3}$$

Q036 Written

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

Solution

1. Factor each polynomial: $\frac{(x-8)(x-3)}{(x-8)(x+2)}$ and $\frac{x(x+2)}{(x-3)^2}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x}{x-3}$.
4. Retain the original denominator restrictions.

Final answer

$$\frac{x}{x-3}$$

Worked Solutions

Rational Expressions

Q037 Written

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}.$

Solution

1. Factor each polynomial: $\frac{x+1}{(x-2)(x+2)}$ and $\frac{(x-4)(x+2)}{x(x+1)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{x-4}{x(x-2)}.$
4. Retain the original denominator restrictions.

Final answer

$$\frac{x-4}{x(x-2)}$$

Q038 Written

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}.$

Solution

1. Factor each polynomial: $\frac{(x-3)(x+3)}{x(x-2)}$ and $\frac{(x-5)(x-2)}{(x-3)(x+4)}$.
2. Multiply the numerators and denominators, then cancel common factors across the product.
3. Therefore the simplified expression is $= \frac{(x-5)(x+3)}{x(x+4)}.$
4. Retain the original denominator restrictions.

Final answer

$$\frac{(x-5)(x+3)}{x(x+4)}$$

Worked Solutions

Rational Expressions

Q039 Written

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

Solution

- Factor each polynomial: $\frac{(x-2)(x+3)}{(x+2)(2x-3)}$ and $\frac{(x+2)(x+4)}{(x-2)(3x+1)}$.
- Multiply the numerators and denominators, then cancel common factors across the product.
- Therefore the simplified expression is $= \frac{(x+3)(x+4)}{(2x-3)(3x+1)}$.
- Retain the original denominator restrictions.

Final answer

$$\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$$

Q040 Written

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

Solution

- Change division to multiplication by the reciprocal of the second fraction.
- Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+1)}{(x+2)(2x+1)}$ and $\frac{(x-3)(2x+1)}{(x-5)(x+4)}$.
- Cancel only common factors; the result is $= \frac{(x-3)(x+1)}{(x+2)(x+4)}$.
- Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(x+1)}{(x+2)(x+4)}$$

Worked Solutions

Rational Expressions

Q041 Written

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-3)(x+3)}{(x-2)(x+1)}$ and $\frac{x+1}{(x+3)(2x-1)}$.
3. Cancel only common factors; the result is $= \frac{x-3}{(x-2)(2x-1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{x-3}{(x-2)(2x-1)}$$

Q042 Written

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{2xy}{3ab}$ and $\frac{9a^2b^3}{8x^3y}$.
3. Cancel only common factors; the result is $= \frac{3ab^2}{4x^2}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3ab^2}{4x^2}$$

Worked Solutions

Rational Expressions

Q043 Written

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x+2)(2x-5)}{(x-3)(x+1)}$ and $\frac{(x-3)^2}{(x-1)(x+2)}$.
3. Cancel only common factors; the result is $= \frac{(x-3)(2x-5)}{(x-1)(x+1)}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{(x-3)(2x-5)}{(x-1)(x+1)}$$

Q044 Written

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-5)(x+2)}{(x+2)(2x-1)}$ and $\frac{3(2x-1)}{(x-5)(x+3)}$.
3. Cancel only common factors; the result is $= \frac{3}{x+3}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{3}{x+3}$$

Worked Solutions

Rational Expressions

Q045 Written

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}$.

Solution

1. Change division to multiplication by the reciprocal of the second fraction.
2. Factor before cancelling, then multiply by the reciprocal: $\frac{(x-y)(x+y)}{(x-y)^2}$ and $\frac{x-y}{x(x+y)}$.
3. Cancel only common factors; the result is $= \frac{1}{x}$.
4. Retain restrictions from both fractions and require the divisor to be non-zero.

Final answer

$$\frac{1}{x}$$

Q046 Written

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ for $x = \sqrt{123}$.

Solution

1. Factor every polynomial and postpone substitution until the symbolic cancellation is complete.
2. The chain simplifies to $= \frac{(x-1)(x^4+x^3+x^2+x+1)}{2}$.
3. Substitute $x = \sqrt{123}$ only after simplifying: $= -\frac{1}{2} + \frac{15129\sqrt{123}}{2}$.
4. Check all original denominators and divisor-nonzero conditions.

Final answer

$$-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$$

Worked Solutions

Rational Expressions

Q047 Challenge

Simplify $\left(\frac{x+1}{x-2} - \frac{2}{x+3} + \frac{3}{x+3} - \frac{1}{x-2}\right)$; state every excluded value.**Solution**

1. Group like denominators, then combine over the two-factor LCM.
2. The reduced expression is $= \frac{x^2 + 4x - 2}{(x-2)(x+3)}$.
3. The original denominators still control the domain.

Final answer

$$\frac{x^2 + 4x - 2}{(x-2)(x+3)}$$

Worked Solutions

Rational Expressions

Q048 Written

Simplify

$$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= x - 1$.
4. Retain all values excluded by the original denominators.

Final answer

$$x - 1$$

Worked Solutions

Rational Expressions

Q049 MCQ

Simplify

$$\frac{x^2 + 2x + 3}{x + 1} + \frac{2x - 1}{x + 1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x^2 + 4x + 2}{x + 1}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{x^2 + 4x + 2}{x + 1}$$

Worked Solutions

Rational Expressions

Q050 MCQ

Simplify

$$\frac{2x^2 - 5}{x^2 - 4} + \frac{x + 6}{x^2 - 4}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^2 + x + 1}{(x - 2)(x + 2)}$.
5. Keep every value excluded by an original denominator.

Correct option: D**Final answer**

$$\frac{2x^2 + x + 1}{(x - 2)(x + 2)}$$

Worked Solutions

Rational Expressions

Q051 MCQ

Simplify

$$\frac{3x^2 + x - 2}{(x - 2)(x + 1)} + \frac{4x - 3}{(x - 2)(x + 1)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{3x^2 + 5x - 5}{(x - 2)(x + 1)}$$

Worked Solutions

Rational Expressions

Q052 MCQ

Simplify

$$\frac{x^3 - 1}{x + 2} + \frac{2x^2 + x + 4}{x + 2};$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x^3 + 2x^2 + x + 3}{x + 2}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{x^3 + 2x^2 + x + 3}{x + 2}$$

Worked Solutions

Rational Expressions

Q053 Written

Simplify

$$\frac{3}{x^2 + x} - \frac{-x + 3}{x^2 + x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{1}{x + 1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{1}{x + 1}$$

Worked Solutions

Rational Expressions

Q054 MCQ

Simplify

$$\frac{x^2 + 4x + 4}{x + 2} - \frac{x + 4}{x + 2}.$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{x(x + 3)}{x + 2}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{x(x + 3)}{x + 2}$$

Worked Solutions

Rational Expressions

Q055 MCQ

Simplify

$$\frac{3x^2 - 2x + 1}{x^2 - 1} - \frac{2x - 5}{x^2 - 1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 - 4x + 6}{(x - 1)(x + 1)}$$

Worked Solutions

Rational Expressions

Q056 MCQ

Simplify

$$\frac{x^2 - 6x + 8}{(x - 1)(x + 2)} - \frac{x^2 - 3x + 2}{(x - 1)(x + 2)};$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{3(x - 2)}{(x - 1)(x + 2)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$-\frac{3(x - 2)}{(x - 1)(x + 2)}$$

Worked Solutions

Rational Expressions

Q057 MCQ

Simplify

$$\frac{2x^3 + x}{x - 3} - \frac{x^2 + 7}{x - 3}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^3 - x^2 + x - 7}{x - 3}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{2x^3 - x^2 + x - 7}{x - 3}$$

Worked Solutions

Rational Expressions

Q058 Written

Simplify

$$\frac{x+4}{x^2-2x} - \frac{3}{x^2-3x+2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x(x-1)}$$

Q059 Written**Simplify**

$$\frac{x-1}{x^2+4x+3} + \frac{2}{x^2+5x+6}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x}{(x+1)(x+2)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x}{(x+1)(x+2)}$$

Worked Solutions

Rational Expressions

Q060 MCQ

Simplify

$$\frac{x-1}{(x+1)(x+3)} + \frac{2x+1}{(x+3)(x+5)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 + 7x - 4}{(x+1)(x+3)(x+5)}$$

Worked Solutions

Rational Expressions

Q061 MCQ

Simplify

$$\frac{x+3}{(x+2)(x+4)} + \frac{x-2}{(x+4)(x+6)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{2x^2 + 9x + 14}{(x+2)(x+4)(x+6)}$$

Worked Solutions

Rational Expressions

Q062 MCQ

Simplify

$$\frac{3x + 2}{(x + 1)(x + 4)} + \frac{x + 5}{(x + 4)(x + 7)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$.
5. Keep every value excluded by an original denominator.

Correct option: D**Final answer**

$$\frac{4x^2 + 29x + 19}{(x + 1)(x + 4)(x + 7)}$$

Worked Solutions

Rational Expressions

Q063 MCQ

Simplify

$$\frac{x+1}{(x+2)(x+5)} + \frac{4x-3}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: A**Final answer**

$$\frac{5x^2 + 14x + 2}{(x+2)(x+5)(x+8)}$$

Worked Solutions

Rational Expressions

Q064 MCQ

Simplify

$$\frac{2x + 1}{(x + 3)(x + 6)} + \frac{x + 4}{(x + 6)(x + 9)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$.
5. Keep every value excluded by an original denominator.

Correct option: B**Final answer**

$$\frac{3x^2 + 26x + 21}{(x + 3)(x + 6)(x + 9)}$$

Worked Solutions

Rational Expressions

Q065 MCQ

Simplify

$$\frac{x-2}{(x+1)(x+5)} + \frac{3x+1}{(x+5)(x+8)}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= \frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$\frac{4x^2 + 10x - 15}{(x+1)(x+5)(x+8)}$$

Worked Solutions

Rational Expressions

Q066 Written

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x+2}{x+1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x+2}{x+1}$$

Q067 Written**Simplify**

$$\frac{x+2}{x} - \frac{x-2}{x-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3x-2}{x(x-1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x-2}{x(x-1)}$$

Worked Solutions

Rational Expressions

Q068 Written

Simplify

$$\frac{2}{x-1} + \frac{3}{x^2+5x-6}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x+15}{(x-1)(x+6)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x+15}{(x-1)(x+6)}$$

Worked Solutions

Rational Expressions

Q069 MCQ

Simplify

$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1}:$$

Solution

1. Factor each denominator completely before combining the fractions.
2. Use the least common denominator containing every required factor once.
3. Rewrite each numerator over that denominator, then combine the numerators only.
4. Reduce the resulting numerator: $= -\frac{x+4}{(x+1)(x+3)}$.
5. Keep every value excluded by an original denominator.

Correct option: C**Final answer**

$$-\frac{x+4}{(x+1)(x+3)}$$

Worked Solutions

Rational Expressions

Q070 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x}{(x-2)(x+2)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x}{(x-2)(x+2)}$$

Q071 Written**Simplify**

$$\frac{x+4}{x^2+2x} - \frac{x+5}{x^2+x-2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= -\frac{2}{x(x-1)}.$
4. Retain all values excluded by the original denominators.

Final answer

$$-\frac{2}{x(x-1)}$$

Worked Solutions

Rational Expressions

Q072 Written

Simplify

$$\frac{x+8}{x^2+x-2} - \frac{x+5}{x^2-1}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2}{(x+1)(x+2)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x+1)(x+2)}$$

Q073 Written**Simplify**

$$\frac{x-3}{x^2+x} + \frac{x}{x^2-1}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2x^2 - 4x + 3}{x(x-1)(x+1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2x^2 - 4x + 3}{x(x-1)(x+1)}$$

Worked Solutions

Rational Expressions

Q074 **Written****Simplify**

$$\frac{4x}{x^2 - 1} - \frac{x - 1}{x^2 + x}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3x - 1}{x(x - 1)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3x - 1}{x(x - 1)}$$

Q075 Written**Simplify**

$$\frac{1}{x^2 + 5x + 6} + \frac{1}{x^2 + 7x + 12}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{2}{(x + 2)(x + 4)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{2}{(x + 2)(x + 4)}$$

Worked Solutions

Rational Expressions

Q076 Written

Simplify

$$\frac{x-1}{x^2+3x+2} - \frac{x-3}{x^2+4x+3}$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{3}{(x+2)(x+3)}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{3}{(x+2)(x+3)}$$

Worked Solutions

Rational Expressions

Q077 MCQ

A learner combines

 $\frac{2x-1}{(x-1)(x+2)} - \frac{x+4}{x+2}$. Correct the numerator line, then simplify the expression:

Solution

1. Keep one common denominator; never add the denominators.
2. Put the entire second numerator in parentheses before applying the sign.
3. After factoring and reducing, the correct result is $= -\frac{x^2+x-3}{(x-1)(x+2)}$.
4. Check the original denominator restrictions.

Correct option: B

Final answer

$$-\frac{x^2+x-3}{(x-1)(x+2)}$$

Worked Solutions

Rational Expressions

Q078 **Written****Simplify**

$$\frac{x^3 + 8x}{x^2 - x - 2} - \frac{8}{x - 2}.$$

Solution

1. Factor each denominator and write the least common denominator.
2. Multiply each numerator by its missing factor; preserve the subtraction sign.
3. Combine and factor the numerator. The simplified result is $= \frac{x^2 + 2x + 4}{x + 1}$.
4. Retain all values excluded by the original denominators.

Final answer

$$\frac{x^2 + 2x + 4}{x + 1}$$

Worked Solutions

Rational Expressions

Q079 Challenge

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

Solution

1. Identify the two family triggers before manipulating the expression.
2. Simplify each component inside-out, keeping the original restrictions.
3. Combine the reduced components: $= -\frac{x^2 - 4x + 2}{2(x - 2)}$.
4. Audit every original inner and outer denominator.

Final answer

$$-\frac{x^2 - 4x + 2}{2(x - 2)}$$

Worked Solutions

Rational Expressions

Q080 Written

Simplify

$$\frac{1}{1 - \frac{1}{x-1}}.$$

Solution

1. The inner denominator requires $x-1 \neq 0$.
2. Multiply the outer numerator and denominator by $x-1$.
3. The result is $\frac{x-1}{x-2}$.
4. The outer denominator also excludes $x = 2$.

Final answer

$$\frac{x-1}{x-2}$$

Q081 Written**Simplify**

$$\frac{1}{1 - \frac{x}{x+1}}.$$

Solution

1. The inner denominator requires $x+1 \neq 0$.
2. Clear the inner denominator $x+1$ in the outer fraction.
3. The denominator becomes 1, so the result is $x + 1$.
4. Retain $x \neq -1$ from the original expression.

Final answer

$$x + 1$$

Worked Solutions

Rational Expressions

Q082 MCQ

Simplify

$$\frac{1}{1 - \frac{x+2}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 1$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-x - 1$$

Q083 MCQ

Simplify

$$\frac{1}{1 - \frac{x+3}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 2$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$-x - 2$$

Worked Solutions

Rational Expressions

Q084 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+1}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x}$$

Q085 MCQ

Simplify

$$\frac{1}{1 - \frac{2x + 3}{x + 1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x + 1}{x + 2}$.
4. State every restriction from the original denominators.

Correct option: C**Final answer**

$$-\frac{x + 1}{x + 2}$$

Worked Solutions

Rational Expressions

Q086 MCQ

Simplify

$$\frac{1}{1 - \frac{3x+1}{x+2}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+2}{2x-1}$.
4. State every restriction from the original denominators.

Correct option: D

Final answer

$$-\frac{x+2}{2x-1}$$

Q087 MCQ

Simplify

$$\frac{1}{1 - \frac{x+4}{x+3}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -x - 3$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$-x - 3$$

Worked Solutions

Rational Expressions

Q088 MCQ

Simplify

$$\frac{1}{1 - \frac{2x+5}{x+1}}:$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= -\frac{x+1}{x+4}$.
4. State every restriction from the original denominators.

Correct option: B

Final answer

$$-\frac{x+1}{x+4}$$

Q089 Written**Simplify**

$$\frac{1}{1-x} + \frac{1}{1+x} \div \left(\frac{1}{1-x} - \frac{1}{1+x} \right).$$

Solution

1. Multiply both outer parts by $(1-x)(1+x)$.
2. The numerator pair becomes 2; the denominator pair becomes $2x$.
3. Reduce to $1/x$.
4. The original inner denominators and the outer denominator give $x \neq -1, 0, 1$.

Final answer

$$1/x$$

Worked Solutions

Rational Expressions

Q090 MCQ

Simplify

$$\frac{1}{x+1} + \frac{1}{x+4} \div \left(\frac{1}{x+1} - \frac{1}{x+4} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= \frac{2x+5}{3}$.
4. State every restriction from the original denominators.

Correct option: A

Final answer

$$\frac{2x+5}{3}$$

Q091 MCQ**Simplify**

$$\frac{1}{x+1} + \frac{1}{x+5} \div \left(\frac{1}{x+1} - \frac{1}{x+5} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: $= \frac{x+3}{2}$.
4. State every restriction from the original denominators.

Correct option: A**Final answer**

$$\frac{x+3}{2}$$

Worked Solutions

Rational Expressions

Q092 Written

Simplify

$$\frac{1}{x+2} + \frac{1}{x-2} \div \left(\frac{1}{x+2} - \frac{1}{x-2} \right).$$

Solution

1. Multiply numerator and denominator by $(x+2)(x-2)$.
2. The numerator pair becomes $2x$; the denominator pair becomes -4 .
3. The result is $-x/2$.
4. Keep $x \neq -2$ and $x \neq 2$.

Final answer

$$-\frac{x}{2}$$

Q093 MCQ**Simplify**

$$\frac{1}{x+1} + \frac{1}{x+3} \div \left(\frac{1}{x+1} + \frac{1}{x+3} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: C**Final answer**

1

Worked Solutions

Rational Expressions

Q094 MCQ

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \div \left(\frac{1}{x+2} + \frac{1}{x+5} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: D

Final answer

1

Q095 MCQ**Simplify**

$$\frac{1}{x+3} + \frac{1}{x+6} \div \left(\frac{1}{x+3} + \frac{1}{x+6} \right):$$

Solution

1. Simplify the innermost fraction first.
2. Multiply the outer numerator and denominator by the required inner denominator.
3. Reduce the resulting expression: = 1.
4. State every restriction from the original denominators.

Correct option: B**Final answer**

1

Worked Solutions

Rational Expressions

Q096 Written

Simplify

$$\frac{1}{x+y} + \frac{1}{x-y} \div \left(\frac{1}{x+y} - \frac{1}{x-y} \right).$$

Solution

1. Clear both inner denominators using $(x+y)(x-y)$.
2. The numerator pair becomes $2x$; the denominator pair becomes $-2y$.
3. Reduce to $-x/y$.
4. Keep $y \neq 0$ and $x \neq \pm y$.

Final answer

$$-\frac{x}{y}$$

Q097 Written**Simplify**

$$\frac{1}{1 - \frac{1}{1 - \frac{1}{x}}}.$$

Solution

1. Start with the innermost $1 - 1/x$.
2. Substitute that result into the next reciprocal layer.
3. The outer result reduces to $1 - x$.
4. The original denominators exclude $x = 0$ and $x = 1$.

Final answer

$$1 - x$$

Worked Solutions

Rational Expressions

Q098 Written

Simplify

$$\frac{1}{1 - \frac{1}{1 + \frac{1}{x}}}.$$

Solution

1. Simplify $1 + 1/x$ before entering the next layer.
2. Substitute the result into $1 - 1/(...)$.
3. The final result is $x + 1$.
4. Retain $x \neq 0$ and $x \neq -1$.

Final answer

$$x + 1$$

Q099 Written**Simplify**

$$\left(1 + \frac{x-y}{x+y}\right) \div \left(1 - \frac{x-y}{x+y}\right).$$

Solution

1. Let $u = (x - y)/(x + y)$, then combine $1+u$ and $1-u$.
2. Clear $x+y$ in the resulting numerator and denominator.
3. The reduced result is $\frac{x}{y}$.
4. Require $y \neq 0$ and $x + y \neq 0$.

Final answer

$$\frac{x}{y}$$

Worked Solutions

Rational Expressions

Q100 Written

Simplify

$$\left(x - \frac{2}{x-1}\right) \div \left(\frac{x}{x-1} - 2\right).$$

Solution

1. Combine the polynomial pieces over $x-1$ in the numerator and denominator.
2. Form the outer quotient only after both pieces are simplified.
3. Cancel complete factors to obtain $-(x+1)$.
4. The original denominators retain $x \neq 1$ and $x \neq 2$.

Final answer

$$-(x+1)$$

Q101 Written

Simplify

$$\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right) \div \left(\frac{x+1}{x-1} + \frac{x-1}{x+1}\right).$$

Solution

1. Use $(x-1)(x+1)$ inside both the numerator and denominator.
2. The difference gives $4x$; the sum gives $2(x^2+1)$.
3. The simplified result is $\frac{2x}{x^2+1}$.
4. The original factors require $x \neq \pm 1$.

Final answer

$$\frac{2x}{x^2+1}$$

Worked Solutions

Rational Expressions

Q102 Written

Simplify

$$\frac{1}{1 + \frac{1}{1 + \frac{1}{1+x}}}.$$

Solution

1. Translate the three booster stages as an inside-out reciprocal chain.
2. Simplify the innermost stage $1+1/(1+x)$.
3. Continuing outward gives $\frac{x+2}{2x+3}$.
4. The stated model condition is $x>0$.

Final answer

$$\frac{x+2}{2x+3}$$

Quiz MCQ 1 Concept Quiz · MCQ

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

Solution

1. Factor both polynomials and cancel $x+2$.

Correct option: A

Final answer

$$\frac{x-2}{3x+1}$$

Worked Solutions

Rational Expressions

Quiz MCQ 2 **Concept Quiz · MCQ****Simplify**

$$\frac{x+1}{x-2} + \frac{2}{x-2}:$$

Solution

1. The denominators match, so add the numerators only.

Correct option: A

Final answer

$$\frac{x+3}{x-2}$$

Quiz MCQ 3 **Concept Quiz · MCQ****In**

$\frac{3}{x^2+x} - \frac{3-x}{x^2+x}$, the numerator after the minus sign is... Which of the following is correct?:

Solution

1. Parentheses preserve the sign of every term in the second numerator.

Correct option: A

Final answer

$$3 - (3 - x)$$

Worked Solutions

Rational Expressions

Quiz MCQ 4 Concept Quiz · MCQ

Which statement is safe when simplifying a compound fraction?:

Solution

1. A common multiplier preserves the quotient and exposes the structure.

Correct option: A

Final answer

multiply top and bottom by the same inner denominator

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ and state the original restrictions.

Solution

1. Factor all four polynomials and cancel common factors.
2. Final form: $= \frac{x+2}{x+1}$; retain original denominator exclusions.

Final answer

$$\frac{x+2}{x+1}$$

Worked Solutions

Rational Expressions

Quiz WRITTEN 6 Concept Quiz · Written

Simplify

$\frac{x^2 - 4}{x + 1} + \frac{3}{x + 1}$ and state the original restriction.

Solution

1. Use the common denominator $x+1$ and combine the numerators.
2. Final form: $= x - 1$; $x \neq -1$ remains excluded.

Final answer

$$x - 1$$

Quiz WRITTEN 7 Concept Quiz · Written

Simplify

$\frac{x^2 - y^2}{x^2 - 2xy + y^2} - \frac{x + y}{x - y}$ in x and y .

Solution

1. Factor $x^2 - y^2$, use the common $(x - y)$ structure, and apply the minus sign to $x + y$.
2. Final form: $= 0$.

Final answer

$$0$$

Worked Solutions

Rational Expressions

Quiz WRITTEN 8 Concept Quiz · Written

Explain why $(x^2 - 16)/(x - 4) = x + 4$ is not an identity for every real x .

Solution

1. The cancellation is valid only when $x - 4 \neq 0$.
2. At $x = 4$ the original expression is undefined, so the graph has a hole.

Final answer

$x \neq 4$; removable hole at $(4, 8)$

Answer key

Use the question number shown on each page.

Q001 A
Q002 B
Q003 C
Q004 D
Q005 A
Q006 B
Q007 $\frac{4b^3}{3a^2}$
Q008 $\frac{3y^2}{10x}$
Q009 $\frac{3a}{4b^3}$
Q010 $\frac{1}{4ab^2}$
Q011 B
Q012 A
Q013 A
Q014 B
Q015 A
Q016 B
Q017 C
Q018 D
Q019 A
Q020 B
Q021 C
Q022 $\frac{3c^2}{a}$
Q023 $\frac{4a^2c^3}{b^2}$

Q024 C
Q025 D
Q026 D
Q027 A
Q028 B
Q029 C
Q030 $\frac{x-2}{3x+1}$
Q031 $\frac{2x-5}{x+2}$
Q032 $\frac{x}{x-3}$
Q033 $\frac{x+2}{x-4}$
Q034 $\frac{2(x-y)}{x(x+4y)}$
Q035 $\frac{x^2+x+1}{x+3}$
Q036 $\frac{x}{x-3}$
Q037 $\frac{x-4}{x(x-2)}$
Q038 $\frac{(x-5)(x+3)}{x(x+4)}$
Q039 $\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$
Q040 $\frac{(x-3)(x+1)}{(x+2)(x+4)}$
Q041 $\frac{x-3}{(x-2)(2x-1)}$
Q042 $\frac{3ab^2}{4x^2}$

Q043 $\frac{(x-3)(2x-5)}{(x-1)(x+1)}$
Q044 $\frac{3}{x+3}$
Q045 $\frac{1}{x}$
Q046 $-\frac{1}{2} + \frac{15129\sqrt{123}}{2}$
Q047 $\frac{x^2+4x-2}{(x-2)(x+3)}$
Q048 $x-1$
Q049 C
Q050 D
Q051 A
Q052 B
Q053 $\frac{1}{x+1}$
Q054 A
Q055 B
Q056 C
Q057 C
Q058 $\frac{x+2}{x(x-1)}$
Q059 $\frac{x}{(x+1)(x+2)}$
Q060 B
Q061 C
Q062 D
Q063 A
Q064 B
Q065 C

Q066 $\frac{x+2}{x+1}$
Q067 $\frac{3x-2}{x(x-1)}$
Q068 $\frac{2x+15}{(x-1)(x+6)}$
Q069 C
Q070 $\frac{2x}{(x-2)(x+2)}$
Q071 $-\frac{2}{x(x-1)}$
Q072 $\frac{2}{(x+1)(x+2)}$
Q073 $\frac{2x^2-4x+3}{x(x-1)(x+1)}$
Q074 $\frac{3x-1}{x(x-1)}$
Q075 $\frac{2}{(x+2)(x+4)}$
Q076 $\frac{3}{(x+2)(x+3)}$
Q077 B
Q078 $\frac{x^2+2x+4}{x+1}$
Q079 $-\frac{x^2-4x+2}{2(x-2)}$
Q080 $\frac{x-1}{x-2}$
Q081 $x+1$
Q082 D
Q083 A
Q084 B
Q085 C

Q086 D

Q087 A

Q088 B

Q089 $1/x$

Q090 A

Q091 A

Q092 $-\frac{x}{2}$

Q093 C

Q094 D

Q095 B

Q096 $-\frac{x}{y}$ Q097 $1 - x$ Q098 $x + 1$ Q099 $\frac{x}{y}$ Q100 $-(x + 1)$ Q101 $\frac{2x}{x^2 + 1}$ Q102 $\frac{x + 2}{2x + 3}$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{x + 2}{x + 1}$ Quiz WRITTEN 6 $x - 1$

Quiz WRITTEN 7 0

Quiz WRITTEN 8 $x \neq 4$; removable hole at (4, 8)

Concept 4 | Identities and Proofs

English Student Edition • Focused formulas and guided practice

Formula opener

Proof route

LHS $\longrightarrow$ RHS

Square

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

Difference

$$a^2 - b^2 = (a - b)(a + b)$$

Worked Solutions

Identities and Proofs

Q001 Challenge

$2x^2 + 3x + 1 = (x + 1)(ax + b) + c$. Find **a,b,c**.

Solution

1. Decide that this is a polynomial identity and expand the product.
2. Match the x^2 , x , and constant coefficients.
3. The constants are $a = 2$, $b = 1$, $c = 0$; substitute back.

Final answer

$$a = 2, b = 1, c = 0$$

Worked Solutions

Identities and Proofs

Q002 Written

$$x^2 + 6 = a(x + 1)^2 + b(x + 1) + c$$

Solution

1. Expand the shifted square: the right side is $ax^2 + (2a+b)x + (a+b+c)$.
2. Match coefficients with $x^2 + 0x + 6$: $a = 1$, $2a + b = 0$, $a + b + c = 6$.
3. Solve in order to obtain $a = 1$, $b = -2$, $c = 7$.
4. Substitution reproduces the original identity.

Final answer

$$a = 1, b = -2, c = 7$$

Q003 Written

$$x^2 - x - 3 = a(x - 1)^2 + b(x - 1) + c$$

Solution

1. Expand and collect the right side as $ax^2 + (-2a+b)x + (a-b+c)$.
2. Match with $x^2 - x - 3$ and solve $a = 1$, $b = 1$, $c = -3$.
3. Check all three coefficients in the original identity.

Final answer

$$a = 1, b = 1, c = -3$$

Worked Solutions

Identities and Proofs

Q004 MCQ

For the identity $x^2 + 6x + 5 = a(x + 2)^2 + b(x + 2) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 1$, $b = 2$, $c = -3$.
4. Substitute back to check.

Correct option: C**Final answer** $(1, 2, -3)$

Q005 MCQ

For the identity $2x^2 + 9x + 8 = a(x + 3)^2 + b(x + 3) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 2$, $b = -3$, $c = -1$.
4. Substitute back to check.

Correct option: C**Final answer** $(2, -3, -1)$

Worked Solutions

Identities and Proofs

Q006 MCQ

For the identity $3x^2 + 28x + 65 = a(x + 4)^2 + b(x + 4) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 3$, $b = 4$, $c = 1$.
4. Substitute back to check.

Correct option: D

Final answer

$(3, 4, 1)$

Q007 MCQ

For the identity $x^2 + 5x + 3 = a(x + 5)^2 + b(x + 5) + c$, the ordered tuple (a, b, c) is:

Solution

1. Expand the shifted square and collect descending powers.
2. Match the x^2 , x , and constant coefficients.
3. Read off $a = 1$, $b = -5$, $c = 3$.
4. Substitute back to check.

Correct option: A

Final answer

$(1, -5, 3)$

Worked Solutions

Identities and Proofs

Q008 Written

$$a(x + 1) - b(x + 5) = 8x$$

Solution

1. Collect coefficients: $(a - b)x + (a - 5b) = 8x$.
2. Solve $a - b = 8$ and $a - 5b = 0$.
3. Hence $a = 10$ and $b = 2$; the constant term checks as zero.

Final answer

$$a = 10, b = 2$$

Q009 MCQ

For the identity $a(x + 1) - b(x + 3) = x - 1$, the ordered tuple (a, b) is:**Solution**

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 2, b = 1$.
4. Check both terms.

Correct option: A**Final answer**

$$(2, 1)$$

Worked Solutions

Identities and Proofs

Q010 MCQ

For the identity $a(x + 2) - b(x + 4) = 5x + 14$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 3$, $b = -2$.
4. Check both terms.

Correct option: B

Final answer $(3, -2)$

Q011 MCQ

For the identity $a(x + 3) - b(x + 5) = x - 3$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 4$, $b = 3$.
4. Check both terms.

Correct option: D

Final answer $(4, 3)$

Worked Solutions

Identities and Proofs

Q012 MCQ

For the identity $a(x + 4) - b(x + 6) = 9x + 44$, the ordered tuple (a, b) is:

Solution

1. Collect the coefficient of x and the constant term.
2. Match the two coefficients with the right side.
3. Solve $a = 5$, $b = -4$.
4. Check both terms.

Correct option: D

Final answer

$(5, -4)$

Q013 Written

$$(x + a)(b + 3) = 4x + 12$$

Solution

1. Expand as $(b+3)x+a(b+3)$.
2. Match $b + 3 = 4$ and $a(b + 3) = 12$.
3. Therefore $b = 1$ and $a = 3$; c is not present in the printed identity and is not applicable.

Final answer

$a = 3, b = 1$

Worked Solutions

Identities and Proofs

Q014 MCQ

For the identity $(x + a)(b + 4) = 3x + 6$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+d)x+a(b+d)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 2$, $b = -1$; no extra parameter is present.
4. Substitute back.

Correct option: B

Final answer

$(2, -1)$

Q015 MCQ

For the identity $(x + a)(b + 5) = 5x + 15$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+d)x+a(b+d)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 3$, $b = 0$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(3, 0)$

Worked Solutions

Identities and Proofs

Q016 MCQ

For the identity $(x + a)(b + 6) = 7x + 28$, the ordered tuple (a, b) is:

Solution

1. Expand as $(b+6)x+a(b+6)$.
2. Match the coefficient of x and the constant term.
3. Obtain $a = 4$, $b = 1$; no extra parameter is present.
4. Substitute back.

Correct option: C

Final answer

$(4, 1)$

Q017 Written

$$2x^2 + 5x + 4 = (x + 1)(ax + b) + c$$

Solution

1. Expand the right side to $ax^2+(a+b)x+(b+c)$.
2. Match $(a, a + b, b + c) = (2, 5, 4)$.
3. Solve $a = 2$, $b = 3$, $c = 1$ and check the constant term.

Final answer

$a = 2$, $b = 3$, $c = 1$

Worked Solutions

Identities and Proofs

Q018 Written

$$(ax + b)(x + 1) = 4x^2 + cx + 1$$

Solution

1. Expand: $ax^2 + (a+b)x + b$.
2. Match the x^2 and constant coefficients to get $a = 4$ and $b = 1$.
3. Then $c = a + b = 5$.

Final answer

$$a = 4, b = 1, c = 5$$

Q019 Written

$$2x^2 - 7x + 8 = (x - 3)(ax + b) + c$$

Solution

1. Expand to $ax^2 + (b-3a)x + (-3b+c)$.
2. Match with $(2, -7, 8)$.
3. Solve $a = 2$, $b = -1$, $c = 5$ and substitute back.

Final answer

$$a = 2, b = -1, c = 5$$

Worked Solutions

Identities and Proofs

Q020 MCQ

For the identity $(ax - 1)(x + 1) + c = x^2 - 3$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (ah+b)x + bh + c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 1$ and $c = -2$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B**Final answer** $(1, -2)$

Q021 MCQ

For the identity $(ax + 2)(x + 2) + c = 2x^2 + 6x + 3$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (ah+b)x + bh + c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 2$ and $c = -1$, then verify the middle coefficient.
4. Check the full identity.

Correct option: C**Final answer** $(2, -1)$

Worked Solutions

Identities and Proofs

Q022 MCQ

For the identity $(ax + 3)(x + 3) + c = 3x^2 + 12x + 9$, the ordered tuple (a, c) is:

Solution

1. Expand the product to $ax^2 + (ah+b)x + bh + c$.
2. Match the x^2 , x , and constant coefficients.
3. Use $a = 3$ and $c = 0$, then verify the middle coefficient.
4. Check the full identity.

Correct option: B

Final answer

$(3, 0)$

Q023 Written

$$\frac{3x + 1}{(x + 2)(x - 3)} = \frac{a}{x + 2} + \frac{b}{x - 3}$$

Solution

1. Exclude $x = -2$ and $x = 3$ from the original denominator.
2. Multiply by $(x+2)(x-3)$: $3x + 1 = a(x - 3) + b(x + 2)$.
3. Match coefficients: $a + b = 3$ and $-3a + 2b = 1$, giving $a = 1$, $b = 2$.
4. Substitute a and b back into the rational identity.

Final answer

$a = 1$, $b = 2$

Worked Solutions

Identities and Proofs

Q024 Written

$$\frac{x+8}{(x-2)(x+3)} = \frac{a}{x-2} + \frac{b}{x+3}$$

Solution

1. Exclude $x = 2$ and $x = -3$.
2. Clear denominators: $x + 8 = a(x + 3) + b(x - 2)$.
3. Match $a + b = 1$ and $3a - 2b = 8$, giving $a = 2$, $b = -1$.
4. Substitute and verify.

Final answer

$$a = 2, b = -1$$

Q025 Written

$$\frac{1}{(2x-1)(x+1)} = \frac{a}{2x-1} + \frac{b}{x+1}$$

Solution

1. Exclude $x = 1/2$ and $x = -1$.
2. Clear denominators: $1 = a(x + 1) + b(2x - 1)$.
3. Match $a + 2b = 0$ and $a - b = 1$, giving $a = 2/3$, $b = -1/3$.
4. Verify in the original decomposition.

Final answer

$$a = \frac{2}{3}, b = -\frac{1}{3}$$

Worked Solutions

Identities and Proofs

Q026 Written

$$\frac{a}{x-1} + \frac{b}{x-5} = \frac{5x-1}{(x-1)(x-5)}$$

Solution

1. Exclude $x = 1$ and $x = 5$.
2. Clear denominators: $a(x-5) + b(x-1) = 5x-1$.
3. Match $a+b=5$ and $-5a-b=-1$, giving $a=-1$, $b=6$.
4. Check the original rational identity.

Final answer

$$a = -1, b = 6$$

Q027 Written

$$\frac{3x-8}{3x^2+5x-2} = \frac{a}{x+2} + \frac{b}{3x-1}$$

Solution

1. Factor $3x^2 + 5x - 2 = (x+2)(3x-1)$.
2. Exclude $x = -2$ and $x = 1/3$, then clear denominators.
3. Match $3a+b=3$ and $-a+2b=-8$, giving $a=2$, $b=-3$.
4. Check the factored rational identity.

Final answer

$$a = 2, b = -3$$

Worked Solutions

Identities and Proofs

Q028 Written

$$f(t) = \frac{22t + 15}{8t^2 + 10t + 3} = \frac{a}{2t + 1} + \frac{b}{4t + 3}, \quad t > 0$$

Solution

1. Factor ($8t^2 + 10t + 3 = (2t + 1)(4t + 3)$).
2. The stated domain is ($t > 0$); the algebraic exclusions lie outside it.
3. Clear denominators: ($22t + 15 = a(4t + 3) + b(2t + 1)$).
4. Match ($4a + 2b = 22$) and ($3a + b = 15$) to get ($a = 4, b = 3$).

Final answer

$$a = 4, \quad b = 3$$

Q029 Written

Prove that $(x + y)^2 - (x - y)^2 = 4xy$.**Solution**

1. Expand both conjugate squares on the left.
2. Remove the parentheses: $x^2 + 2xy + y^2 - x^2 + 2xy - y^2$.
3. Collect like terms to obtain $4xy$, which is the right-hand side.

Final answer

$$4xy$$

Worked Solutions

Identities and Proofs

Q030 Written

Prove that $(2a + b)^2 - (a + 2b)^2 = 3(a^2 - b^2)$.**Solution**

1. Recognise a difference of squares: $U^2 - V^2 = (U - V)(U + V)$.
2. Set $U = 2a + b$ and $V = a + 2b$; then $U - V = a - b$ and $U + V = 3a + 3b$.
3. Multiply to get $(a - b)(3a + 3b) = 3(a^2 - b^2)$.

Final answer

$$3a^2 - 3b^2$$

Q031 MCQ

For the identity $-(x - 2y + 1)^2 + (x + 2y + 1)^2 = 2y(4x + 4)$, **the value of the simplified left-hand side is:****Solution**

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy + 8y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A**Final answer**

$$8xy + 8y$$

Worked Solutions

Identities and Proofs

Q032 MCQ

For the identity $-(x - 3y + 2)^2 + (x + 3y + 2)^2 = 3y(4x + 8)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $12xy + 24y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer** $12xy + 24y$

Q033 MCQ

For the identity $-(x - y + 3)^2 + (x + y + 3)^2 = y(4x + 12)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $4xy + 12y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer** $4xy + 12y$

Worked Solutions

Identities and Proofs

Q034 MCQ

For the identity $-(x - 2y + 4)^2 + (x + 2y + 4)^2 = 2y(4x + 16)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy + 32y$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: A

Final answer

$$8xy + 32y$$

Q035 Written

Prove that $(x + y)^2 + (x - y)^2 = 2x^2 + 2y^2$.

Solution

1. Expand both squares on the left.
2. Add $x^2 + 2xy + y^2$ and $x^2 - 2xy + y^2$.
3. The mixed terms cancel, leaving $2x^2 + 2y^2$.

Final answer

$$2x^2 + 2y^2$$

Worked Solutions

Identities and Proofs

Q036 MCQ

For the identity $(x - 2y + 1)^2 + (x + 2y + 1)^2 = 8y^2 + 2(x + 1)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 4x + 8y^2 + 2$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$2x^2 + 4x + 8y^2 + 2$$

Q037 MCQ

For the identity $(x - 3y + 2)^2 + (x + 3y + 2)^2 = 18y^2 + 2(x + 2)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 8x + 18y^2 + 8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$2x^2 + 8x + 18y^2 + 8$$

Worked Solutions

Identities and Proofs

Q038 MCQ

For the identity $(x - y + 3)^2 + (x + y + 3)^2 = 2y^2 + 2(x + 3)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 12x + 2y^2 + 18$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C

Final answer

$$2x^2 + 12x + 2y^2 + 18$$

Q039 MCQ

For the identity $(x - 2y + 4)^2 + (x + 2y + 4)^2 = 8y^2 + 2(x + 4)^2$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $2x^2 + 16x + 8y^2 + 32$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D

Final answer

$$2x^2 + 16x + 8y^2 + 32$$

Worked Solutions

Identities and Proofs

Q040 Written

Prove that $(a + 2b)^2 - 4ab = (a - 2b)^2 + 4ab$.**Solution**

1. Expand the left-hand side: $a^2 + 4ab + 4b^2 - 4ab = a^2 + 4b^2$.
2. Expand the right-hand side: $a^2 - 4ab + 4b^2 + 4ab = a^2 + 4b^2$.
3. Both sides simplify to the same expression.

Final answer

$$a^2 + 4b^2$$

Q041 MCQ

For the identity $-(x - 2y + 3)^2 + (x + 2y - 1)^2 = (8x + 8)(y - 1)$, **the value of the simplified left-hand side is:****Solution**

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $8xy - 8x + 8y - 8$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: D**Final answer**

$$8xy - 8x + 8y - 8$$

Worked Solutions

Identities and Proofs

Q042 MCQ

For the identity $-(2x - 3y + 5)^2 + (2x + 3y - 1)^2 = (24x + 24)(y - 1)$, the value of the simplified left-hand side is:

Solution

1. Set U and V equal to the two repeated algebraic parts.
2. Use the matching conjugate-square identity and expand only if needed.
3. The expression reduces to $24xy - 24x + 24y - 24$.
4. The two sides are therefore equal for all real values of the variables.

Correct option: C**Final answer**

$$24xy - 24x + 24y - 24$$

Q043 Written

Prove that $(x^2 + x + 1)(x^2 - x + 1) = x^4 + x^2 + 1$.

Solution

1. Group the factors as $[(x^2 + 1) - x][(x^2 + 1) + x]$.
2. Use $(U - V)(U + V) = U^2 - V^2$ with $U = x^2 + 1$ and $V = x$.
3. Simplify $(x^2 + 1)^2 - x^2$ to $x^4 + x^2 + 1$.

Final answer

$$x^4 + x^2 + 1$$

Worked Solutions

Identities and Proofs

Q044 Written

Prove that $(a^2 - ab + b^2)(a^2 + ab + b^2) = (a^2 + b^2)^2 - (ab)^2$.

Solution

1. Group the factors as $[(a^2 + b^2) - ab][(a^2 + b^2) + ab]$.
2. Apply $(U - V)(U + V) = U^2 - V^2$ with $U = a^2 + b^2$ and $V = ab$.
3. This gives $(a^2 + b^2)^2 - (ab)^2$, exactly the right-hand side.

Final answer

$$a^4 + a^2b^2 + b^4$$

Q045 Written

Prove that $(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2$.

Solution

1. Expand the left-hand product to $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$.
2. Expand the first right-hand square: $a^2c^2 - 2abcd + b^2d^2$.
3. Expand the second right-hand square: $a^2d^2 + 2abcd + b^2c^2$.
4. Add the two expansions; the $-2abcd$ and $+2abcd$ terms cancel, leaving the left-hand expression.

Final answer

$$a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$$

Worked Solutions

Identities and Proofs

Q046 Written

Prove that $(x^2 + 1)(y^2 + 1) = (xy + 1)^2 + (x - y)^2$.

Solution

1. Expand the left-hand side to $x^2y^2 + x^2 + y^2 + 1$.
2. Expand the right-hand side: $x^2y^2 + 2xy + 1 + x^2 - 2xy + y^2$.
3. The $+2xy$ and $-2xy$ terms cancel, leaving the same expression as the left.

Final answer

$$x^2y^2 + x^2 + y^2 + 1$$

Q047 Written

Prove that $(a^2 - b^2)(c^2 - d^2) = (ac - bd)^2 - (ad - bc)^2$.

Solution

1. Expand the first right-hand square: $a^2c^2 - 2abcd + b^2d^2$.
2. Expand the second right-hand square: $a^2d^2 - 2abcd + b^2c^2$.
3. Subtract; the mixed terms cancel, giving $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$.
4. Factor the result as $(a^2 - b^2)(c^2 - d^2)$.

Final answer

$$a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$$

Worked Solutions

Identities and Proofs

Q048 Written

Prove that $(a + b)^3 - 3ab(a + b) = a^3 + b^3$.**Solution**

1. Expand $(a + b)^3$ to $a^3 + 3a^2b + 3ab^2 + b^3$.
2. Expand $3ab(a+b)$ to $3a^2b + 3ab^2$.
3. Subtract the matching cross terms; the result is $a^3 + b^3$.

Final answer

$$a^3 + b^3$$

Q049 Written

Prove that $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = a^3 + b^3 + c^3 - 3abc$.**Solution**

1. Distribute $a+b+c$ across the second bracket.
2. Write all products in a fixed order: a^3, b^3, c^3 , then the mixed terms.
3. Each mixed term cancels with its opposite, while $-abc$ appears three times.
4. The remaining expression is $a^3 + b^3 + c^3 - 3abc$.

Final answer

$$a^3 - 3abc + b^3 + c^3$$

Worked Solutions

Identities and Proofs

Q050 Challenge

For the integrated identity $(x^2 + 1)(-(x - 2y + 1)^2 + (x + 2y + 1)^2) = 2y(4x + 4)(x^2 + 1)$, the value of the simplified left-hand side is:

Solution

1. Recognise the cubic or symmetric pattern before expanding.
2. Expand in descending degree and collect like terms.
3. The cross terms cancel, leaving $8x^3y + 8x^2y + 8xy + 8y$.
4. Therefore the proposed equality is proved.

Correct option: B

Final answer

$$8x^3y + 8x^2y + 8xy + 8y$$

Worked Solutions

Identities and Proofs

Q051 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 + 1 = 2(x + y - xy)$.

Solution

1. Use $x + y = 1$, so $(x + y)^2 = 1$.
2. Expand: $x^2 + 2xy + y^2 = 1$, hence $x^2 + y^2 + 1 = 2 - 2xy$.
3. Since $x + y = 1$, $2 - 2xy = 2(x + y - xy)$. Therefore the equality holds.

Final answer

$$x^2 + y^2 + 1 = 2(x + y - xy)$$

Q052 Written

Under the condition $x + y = 1$, prove that $x^2 + y^2 = 1 - 2xy$.

Solution

1. Square the condition: $(x + y)^2 = 1$.
2. Expand to $x^2 + 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 1 - 2xy$.

Final answer

$$x^2 + y^2 = 1 - 2xy$$

Worked Solutions

Identities and Proofs

Q053 MCQ

Under the condition $x + y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 2$.
2. $y = 2 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$-2xy + 4$$

Q054 MCQ

Under the condition $x + y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 3$.
2. $y = 3 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$$-2xy + 9$$

Worked Solutions

Identities and Proofs

Q055 MCQ

Under the condition $x + y = -1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = -1$.
2. $y = -1 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$-2xy + 1$$

Q056 MCQ

Under the condition $x + y = 4$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x + y = 4$.
2. $y = 4 - x$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $-2xy + 16$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: A

Final answer

$$-2xy + 16$$

Worked Solutions

Identities and Proofs

Q057 Written

Under the condition $x - y = 1$, prove that $x^2 + y^2 = 2xy + 1$.

Solution

1. Square the condition: $(x - y)^2 = 1$.
2. Expand to $x^2 - 2xy + y^2 = 1$.
3. Rearrange to obtain $x^2 + y^2 = 2xy + 1$.

Final answer

$$x^2 + y^2 = 2xy + 1$$

Q058 MCQ

Under the condition $x - y = 1$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 1$.
2. $x = y + 1$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 1$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B**Final answer**

$$2xy + 1$$

Worked Solutions

Identities and Proofs

Q059 MCQ

Under the condition $x - y = 2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 2$.
2. $x = y + 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: B

Final answer

$$2xy + 4$$

Q060 MCQ

Under the condition $x - y = -2$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = -2$.
2. $x = y - 2$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 4$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: C

Final answer

$$2xy + 4$$

Worked Solutions

Identities and Proofs

Q061 MCQ

Under the condition $x - y = 3$, the expression $x^2 + y^2$ is equal to:

Solution

1. Use the condition $x - y = 3$.
2. $x = y + 3$ and substitute into the left-hand side.
3. Expand and collect like terms to obtain $2xy + 9$.
4. This is the stated right-hand side, so the equality is proved.

Correct option: D

Final answer

$$2xy + 9$$

Q062 Written

Under the condition $x + y = 1$, prove that $x^2 - x = y^2 - y$.

Solution

1. Move the right-hand side to the left: $x^2 - y^2 - x + y$.
2. Factor: $x^2 - y^2 - x + y = (x - y)(x + y - 1)$.
3. The condition makes $x + y - 1 = 0$, so the difference is zero.

Final answer

$$x^2 - x = y^2 - y$$

Worked Solutions

Identities and Proofs

Q063 Written

Under the condition $a + b + c = 0$, prove that $a^2 + b^2 + c^2 = -2(ab + bc + ca)$.

Solution

1. From $a + b + c = 0$, write $c = -a - b$.
2. Substitute into the left-hand side: $a^2 + b^2 + (-a - b)^2 = 2a^2 + 2ab + 2b^2$.
3. Substitute into the right-hand side: $-2[ab + b(-a - b) + (-a - b)a] = 2a^2 + 2ab + 2b^2$.
4. Both sides reduce to the same expression, so the equality is proved.

Final answer

$$a^2 + b^2 + c^2 = -2(ab + bc + ca)$$

Q064 Written

Under the condition $a + b + c = 0$, prove that $a^3 + b^3 + c^3 = 3abc$.

Solution

1. Use the identity $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$.
2. The condition gives $a + b + c = 0$, so the product on the right is zero.
3. Therefore $a^3 + b^3 + c^3 - 3abc = 0$, which proves $a^3 + b^3 + c^3 = 3abc$.

Final answer

$$a^3 + b^3 + c^3 = 3abc$$

Worked Solutions

Identities and Proofs

Q065 Written

Under the condition $a + b + c = 0$, prove that $a^2 + ca = b^2 + bc$.

Solution

1. Subtract the right-hand side: $a^2 + ca - b^2 - bc$.
2. Factor by grouping: $(a - b)(a + b) + (a - b)c = (a - b)(a + b + c)$.
3. Since $a + b + c = 0$, the difference is zero; hence the two sides are equal.

Final answer

$$a^2 + ca = b^2 + bc$$

Q066 Written

Under the condition $a + b + c = 0$, prove that $(a + b)(b + c)(c + a) + abc = 0$.

Solution

1. From $a + b + c = 0$, write $a + b = -c$, $b + c = -a$, and $c + a = -b$.
2. Then $(a + b)(b + c)(c + a) = (-c)(-a)(-b) = -abc$.
3. Adding abc gives zero, so the required equality is proved.

Final answer

$$(a + b)(b + c)(c + a) + abc = 0$$

Worked Solutions

Identities and Proofs

Q067 Written

Under the condition $a + b + c = 0$, $abc \neq 0$, prove that $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$.

Solution

1. Because $a + b + c = 0$, $b + c = -a$, $c + a = -b$, and $a + b = -c$.
2. The non-zero condition ensures that all three denominators are non-zero.
3. Therefore each fraction equals -1: $a/(b + c) = b/(c + a) = c/(a + b) = -1$.
4. Their sum is $-1 - 1 - 1 = -3$.

Final answer

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$$

Worked Solutions

Identities and Proofs

Q068 Challenge

Under the condition $x + y = 3$, the expression $x^2 + y^2 + 1$ is equal to:

Solution

1. Use $x + y = 3$ to replace each cyclic denominator.
2. The non-zero condition keeps every original denominator defined.
3. Each term simplifies and the total becomes $-2xy + 10$.
4. Thus the required constant sum is proved.

Correct option: C

Final answer

$$-2xy + 10$$

Worked Solutions

Identities and Proofs

Q069 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$.

Solution

1. Let $a/b = c/d = k$, so $a = bk$ and $c = dk$.
2. Substitute into the left-hand side: $ab/(a^2+b^2) = k/(k^2+1)$.
3. Substitute into the right-hand side: $cd/(c^2+d^2) = k/(k^2+1)$.
4. The two sides are equal.

Final answer

$$\frac{ab}{a^2 + b^2} = \frac{cd}{c^2 + d^2}$$

Q070 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q071 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Q072 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q073 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q074 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q075 Challenge

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{cd}{c^2+d^2}$.
4. Therefore the required equality is proved.

Correct option: C

Final answer

$$\frac{cd}{c^2 + d^2}$$

Worked Solutions

Identities and Proofs

Q076 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$.

Solution

1. Let $a = bk$ and $c = dk$.
2. The left-hand side becomes $k(b^2 + d^2)/(b^2 - d^2)$.
3. The right-hand side becomes $k^2(b^2 + d^2)/[k^2(b^2 - d^2)]$, which is the same expression.
4. Hence the equality is proved.

Final answer

$$\frac{ab + cd}{ab - cd} = \frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q077 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q078 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q079 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: D

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q080 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{a^2+c^2}{a^2-c^2}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{a^2 + c^2}{a^2 - c^2}$$

Worked Solutions

Identities and Proofs

Q081 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}$, $\frac{a+b}{a-b} = \frac{c+d}{c-d}$.

Solution

1. Write $a = bk$ and $c = dk$.
2. Then $(a + b)/(a - b) = (bk + b)/(bk - b) = (k + 1)/(k - 1)$.
3. Similarly $(c + d)/(c - d) = (dk + d)/(dk - d) = (k + 1)/(k - 1)$.
4. The equality follows.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q082 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions

Identities and Proofs

Q083 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: B

Final answer

$$\frac{c+d}{c-d}$$

Q084 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{c+d}{c-d}$$

Worked Solutions

Identities and Proofs

Q085 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{c+d}{c-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{c+d}{c-d}$$

Q086 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$.

Solution

1. Substitute $a = bk$ and $c = dk$.
2. The left-hand side becomes $(3k + k)/(3 + 1) = k$.
3. The right-hand side becomes $(2k - k)/(2 - 1) = k$.
4. Thus both sides are equal.

Final answer

$$\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$$

Worked Solutions

Identities and Proofs

Q087 MCQ

Under the condition $a/b = c/d$, the required expression is equal to:

Solution

1. Use the condition $a/b = c/d$.
2. Introduce one common ratio k and substitute the proportional variables.
3. Simplify both sides to $\frac{2a-c}{2b-d}$.
4. Therefore the required equality is proved.

Correct option: A

Final answer

$$\frac{2a - c}{2b - d}$$

Q088 Written

Under the condition $x/a = y/b$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b}, \quad \frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$.

Solution

1. Let $x = ak$ and $y = bk$.
2. The left-hand side is $(a^2 + a^2k^2)/(a^2 - a^2k^2) = (1 + k^2)/(1 - k^2)$.
3. The right-hand side reduces to the same value.
4. Hence the equality is proved.

Final answer

$$\frac{a^2 + x^2}{a^2 - x^2} = \frac{b^2 + y^2}{b^2 - y^2}$$

Worked Solutions

Identities and Proofs

Q089 Written

Under the condition $a/b = c/d$, prove or find the required expression: $\frac{a}{b} = \frac{c}{d}, \quad \frac{a+b}{a-b} = \frac{c+d}{c-d}.$

Solution

1. Use the equal concentration condition to write $a = bk$ and $c = dk$.
2. The first ratio becomes $(k+1)/(k-1)$.
3. The second ratio becomes $(k+1)/(k-1)$.
4. The required equality follows, provided the denominators are non-zero.

Final answer

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

Q090 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}.$

Solution

1. Let the common ratio be k , so $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also $xy + yz + zx = k^2(ab + bc + ca)$.
4. Both ratios therefore equal k^2 .

Final answer

$$\frac{x^2 + y^2 + z^2}{a^2 + b^2 + c^2} = \frac{xy + yz + zx}{ab + bc + ca}$$

Worked Solutions

Identities and Proofs

Q091 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x+y+z}{a+b+c} = \frac{x}{a}.$

Solution

1. Let the common ratio be k .
2. Then $x + y + z = k(a + b + c)$.
3. Therefore $(x + y + z)/(a + b + c) = k = x/a$.

Final answer

$$\frac{x + y + z}{a + b + c} = \frac{x}{a}$$

Q092 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}, \quad \frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}.$

Solution

1. Set $x = ak$, $y = bk$ and $z = ck$.
2. The left numerator is $k(a+2b+3c)$, so the left ratio is k .
3. The right ratio is also k because $x + y + z = k(a + b + c)$.
4. The two ratios are equal.

Final answer

$$\frac{x + 2y + 3z}{a + 2b + 3c} = \frac{x + y + z}{a + b + c}$$

Worked Solutions

Identities and Proofs

Q093 Written

Under the condition $x : y : z = 2 : 3 : 5$, prove or find the required expression: $x : y : z = 2 : 3 : 5$, $\frac{2xy+yz+3zx}{x^2+y^2+z^2}$.

Solution

1. Let $x = 2k$, $y = 3k$ and $z = 5k$.
2. The numerator is $(12 + 15 + 30)k^2 = 57k^2$.
3. The denominator is $(4 + 9 + 25)k^2 = 38k^2$.
4. Therefore the value is $57/38$.

Final answer

$$\frac{57}{38}$$

Q094 Written

Under the condition $a : b : c = 2 : 3 : 4$, prove or find the required expression: $a : b : c = 2 : 3 : 4$, $\frac{ab+bc+ca}{a^2+b^2+c^2}$.

Solution

1. Let $a = 2k$, $b = 3k$ and $c = 4k$.
2. The numerator is $(6 + 12 + 8)k^2 = 26k^2$.
3. The denominator is $(4 + 9 + 16)k^2 = 29k^2$.
4. Therefore the value is $26/29$.

Final answer

$$\frac{26}{29}$$

Worked Solutions

Identities and Proofs

Q095 Written

Under the condition $a : b : c = 3 : 5 : 2$, prove or find the required expression: $a : b : c = 3 : 5 : 2$, $\frac{a^2+b^2-c^2}{a^2-b^2+c^2}$.

Solution

1. Let $a = 3k$, $b = 5k$ and $c = 2k$.
2. The numerator is $(9 + 25 - 4)k^2 = 30k^2$.
3. The denominator is $(9 - 25 + 4)k^2 = -12k^2$.
4. Therefore the value is $-5/2$.

Final answer

$$-\frac{5}{2}$$

Q096 Written

Under the condition $x : y : z = 2 : -3 : 4$, prove or find the required expression: $x : y : z = 2 : -3 : 4$, $\frac{x^2+y^2+z^2}{xy+yz+zx}$.

Solution

1. Let $x = 2k$, $y = -3k$ and $z = 4k$.
2. The numerator is $(4 + 9 + 16)k^2 = 29k^2$.
3. The denominator is $(-6 - 12 + 8)k^2 = -10k^2$.
4. Therefore the value is $-29/10$.

Final answer

$$-\frac{29}{10}$$

Worked Solutions

Identities and Proofs

Q097 Written

Under the condition $x/7 = y/3 \neq 0$, prove or find the required expression: $\frac{x}{7} = \frac{y}{3} \neq 0$, $\frac{x^2+5y^2}{x^2+xy+y^2}$.

Solution

1. Let $x = 7k$ and $y = 3k$, where k is non-zero.
2. The numerator is $(49 + 45)k^2 = 94k^2$.
3. The denominator is $(49 + 21 + 9)k^2 = 79k^2$.
4. Therefore the value is $94/79$.

Final answer

$$\frac{94}{79}$$

Q098 Written

Under the condition $x/a = y/b = z/c$, prove or find the required expression: $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Solution

1. Let $x = ak$, $y = bk$ and $z = ck$.
2. Then $x^2 + y^2 + z^2 = k^2(a^2 + b^2 + c^2)$.
3. Also $ax + by + cz = k(a^2 + b^2 + c^2)$.
4. Both sides equal $k^2(a^2 + b^2 + c^2)^2$.

Final answer

$$(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$$

Worked Solutions

Identities and Proofs

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x^2 + 4x + 1 = (x + 1)(ax + b) + c$, then (a, b, c) is... Which of the following is correct?:

Solution

1. Expand to $ax^2 + (a + b)x + b + c$.
2. Match $a = 1$, $a + b = 4$, $b + c = 1$, so $b = 3$, $c = -2$.

Correct option: A

Final answer

$(1, 3, -2)$

Quiz MCQ 2 **Concept Quiz · MCQ**

For the identity $(x^2 - 2x + 2)(x^2 + 2x + 2) = -4x^2 + (x^2 + 2)^2$, the value of the simplified left-hand side is:

Solution

1. Rewrite the factors as $(U-V)(U+V)$.
2. Apply $U^2 - V^2$ before carrying out any long expansion.
3. Simplify to $x^4 + 4$.
4. This is exactly the stated right-hand side.

Correct option: A

Final answer

$x^4 + 4$

Worked Solutions

Identities and Proofs

Quiz MCQ 3 Concept Quiz · MCQ

Under the condition $a + b + c = 0$, the expression $3a^2 + 3ac$ is equal to:

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $3b^2 + 3bc$.
4. Hence the two sides are equal under the given condition.

Correct option: A

Final answer

$$3b^2 + 3bc$$

Quiz MCQ 4 Concept Quiz · MCQ

Under the condition $x : y : z = 2 : 3 : 5$, the required expression is equal to:

Solution

1. Write the variables as the stated multiples of k.
2. Substitute into the numerator and denominator.
3. Cancel the common factor and simplify to $\frac{3}{2}$.
4. Hence the required value is obtained.

Correct option: B

Final answer

$$\frac{3}{2}$$

Worked Solutions

Identities and Proofs

Quiz WRITTEN 5 Concept Quiz · Written

Find a, b, c if $x^2 - 3x + 5 = (x - 2)(ax + b) + c$.

Solution

1. Expand to $ax^2 + (b - 2a)x - 2b + c$.
2. Match $a = 1$, $b - 2 = -3$, $-2b + c = 5$.
3. Therefore $b = -1$, $c = 3$.

Final answer

$$a = 1, b = -1, c = 3$$

Quiz WRITTEN 6 Concept Quiz · Written

Prove that $(x^2 - 3x + 4)(x^2 + 3x + 4) = -9x^2 + (x^2 + 4)^2$.

Solution

1. Rewrite the factors as $(U - V)(U + V)$.
2. Apply $U^2 - V^2$ before carrying out any long expansion.
3. Simplify to $x^4 - x^2 + 16$.
4. This is exactly the stated right-hand side.

Final answer

$$x^4 - x^2 + 16$$

Worked Solutions

Identities and Proofs

Quiz WRITTEN 7 Concept Quiz · Written

Under the condition $a + b + c = 0$, prove that $4a^2 + 4ac = 4b^2 + 4bc$.

Solution

1. Move the right-hand side to the left and factor the difference.
2. The factor containing the condition $a + b + c = 0$ is zero.
3. The remaining product is therefore 0, giving $4b^2 + 4bc$.
4. Hence the two sides are equal under the given condition.

Final answer

$$4b^2 + 4bc$$

Quiz WRITTEN 8 Concept Quiz · Written

Under the condition $x/a = y/b = z/c$, prove that $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$.

Solution

1. Use the common multiplier from $x/a = y/b = z/c$.
2. Substitute and collect the common factor.
3. Both sides simplify to $(ax + by + cz)^2$.
4. Thus the identity is proved.

Final answer

$$(ax + by + cz)^2$$

Answer key

Use the question number shown on each page.

Q001 $a = 2, b = 1, c = 0$

Q002 $a = 1, b = -2, c = 7$

Q003 $a = 1, b = 1, c = -3$

Q004 C

Q005 C

Q006 D

Q007 A

Q008 $a = 10, b = 2$

Q009 A

Q010 B

Q011 D

Q012 D

Q013 $a = 3, b = 1$

Q014 B

Q015 C

Q016 C

Q017 $a = 2, b = 3, c = 1$

Q018 $a = 4, b = 1, c = 5$

Q019 $a = 2, b = -1, c = 5$

Q020 B

Q021 C

Q022 B

Q023 $a = 1, b = 2$

Q024 $a = 2, b = -1$

Q025 $a = \frac{2}{3}, b = -\frac{1}{3}$

Q026 $a = -1, b = 6$

Q027 $a = 2, b = -3$

Q028 $a = 4, b = 3$

Q029 $4xy$

Q030 $3a^2 - 3b^2$

Q031 A

Q032 D

Q033 D

Q034 A

Q035 $2x^2 + 2y^2$

Q036 C

Q037 C

Q038 C

Q039 D

Q040 $a^2 + 4b^2$

Q041 D

Q042 C

Q043 $x^4 + x^2 + 1$

Q044 $a^4 + a^2b^2 + b^4$

Q045 $a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$

Q046 $x^2y^2 + x^2 + y^2 + 1$

Q047 $a^2c^2 - a^2d^2 - b^2c^2 + b^2d^2$

Q048 $a^3 + b^3$

Q049 $a^3 - 3abc + b^3 + c^3$

Q050 B

Q051 $x^2 + y^2 + 1 = 2(x + y - xy)$

Q052 $x^2 + y^2 = 1 - 2xy$

Q053 C

Q054 D

Q055 C

Q056 A

Q057 $x^2 + y^2 = 2xy + 1$

Q058 B

Q059 B

Q060 C

Q061 D

Q062 $x^2 - x = y^2 - y$

Q063 $a^2 + b^2 + c^2 = -2(ab + bc + ca)$

Q064 $a^3 + b^3 + c^3 = 3abc$

Q065 $a^2 + ca = b^2 + bc$

Q066 $(a + b)(b + c)(c + a) + abc = 0$

Q067 $\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} = -3$

Q068 C

Q069 $\frac{ab}{a^2+b^2} = \frac{cd}{c^2+d^2}$

Q070 C

Q071 D

Q072 C

Q073 D

Q074 D

Q075 C

Q076 $\frac{ab+cd}{ab-cd} = \frac{a^2+c^2}{a^2-c^2}$

Q077 A

Q078 D

Q079 D

Q080 A

Q081 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q082 B

Q083 B

Q084 A

Q085 A

Q086 $\frac{3a+c}{3b+d} = \frac{2a-c}{2b-d}$

Q087 A

Q088 $\frac{a^2+x^2}{a^2-x^2} = \frac{b^2+y^2}{b^2-y^2}$

Q089 $\frac{a+b}{a-b} = \frac{c+d}{c-d}$

Q090 $\frac{x^2+y^2+z^2}{a^2+b^2+c^2} = \frac{xy+yz+zx}{ab+bc+ca}$

Q091 $\frac{x+y+z}{a+b+c} = \frac{x}{a}$

Q092 $\frac{x+2y+3z}{a+2b+3c} = \frac{x+y+z}{a+b+c}$

Q093 $\frac{57}{38}$

Q094 $\frac{26}{29}$

Q095 $-\frac{5}{2}$

Q096 $-\frac{29}{10}$

Q097 $\frac{94}{79}$

Q098 $(a^2 + b^2 + c^2)(x^2 + y^2 + z^2) = (ax + by + cz)^2$

Quiz MCQ 1 A**Quiz MCQ 2** A**Quiz MCQ 3** A**Quiz MCQ 4** B**Quiz WRITTEN 5** $a = 1, b = -1, c = 3$ **Quiz WRITTEN 6** $x^4 - x^2 + 16$ **Quiz WRITTEN 7** $4b^2 + 4bc$ **Quiz WRITTEN 8** $(ax + by + cz)^2$

Concept 5 | Inequalities

English Student Edition • Focused formulas and guided practice

Formula opener

Square rule

$$(x - y)^2 \geq 0$$

Order

$$0 < a < b \Rightarrow a^2 < b^2$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Worked Solutions

Inequalities

Q001 Written

Under the condition $a > b$, $x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

Solution

1. Start with LHS-RHS: $2(ax+by)-(a+b)(x+y)$.
2. Expand and factor: $ax-ay-bx+by=(a-b)(x-y)$.
3. Both factors are positive, so $(a-b)(x-y)>0$; hence the required inequality holds.

Final answer

$$2(ax + by) > (a + b)(x + y)$$

Q002 MCQ

If $x>y$, prove that $3x + y > 2(x + y)$.

If $x>y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x>y$ makes each factor positive.
4. Hence $3x + y > 2(x + y)$.

Correct option: A

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q003 MCQ

If $x > y$, prove that $3x + 2y > 2x + 3y$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 2y > 2x + 3y$.

Correct option: C

Final answer

$$x - y$$

Q004 MCQ

If $x > y$, prove that $3(x + y) > 2(x + 2y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3(x + y) > 2(x + 2y)$.

Correct option: D

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q005 MCQ

If $x > y$, prove that $3x + 4y > 2x + 5y$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 4y > 2x + 5y$.

Correct option: D

Final answer

$$x - y$$

Q006 MCQ

If $x > y$, prove that $3x + 5y > 2(x + 3y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 5y > 2(x + 3y)$.

Correct option: B

Final answer

$$x - y$$

Worked Solutions

Inequalities

Q007 Written

Under the condition $a > b$, $c > d$, prove that $ac + bd > ad + bc$.

Solution

1. Subtract the right-hand side: $ac + bd - ad - bc$.
2. Factor: $a(c - d) - b(c - d) = (a - b)(c - d)$.
3. Both factors are positive, so the difference is positive; therefore $ac + bd > ad + bc$.

Final answer

$$ac + bd > ad + bc$$

Q008 Written

Under the condition $a > c$, $b > d$, prove that $ab + cd > ad + bc$.

Solution

1. Subtract the right-hand side: $ab + cd - ad - bc$.
2. Factor: $a(b - d) - c(b - d) = (a - c)(b - d)$.
3. Both factors are positive, so the difference is positive; hence $ab + cd > ad + bc$.

Final answer

$$ab + cd > ad + bc$$

Worked Solutions

Inequalities

Q009 MCQ

If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$.

If $a > b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - b)(c - d)$.
2. Factor the difference as $(a - b)(c - d)$.
3. The condition $a > b$ and $c > d$ makes each factor positive.
4. Hence $ac + bd > ad + bc$.

Correct option: D

Final answer

$$(a - b)(c - d)$$

Q010 MCQ

If $a > 2b$ and $c > d$, prove that $ac + 2bd > ad + 2bc$.

If $a > 2b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 2b)(c - d)$.
2. Factor the difference as $(a - 2b)(c - d)$.
3. The condition $a > 2b$ and $c > d$ makes each factor positive.
4. Hence $ac + 2bd > ad + 2bc$.

Correct option: C

Final answer

$$(a - 2b)(c - d)$$

Worked Solutions

Inequalities

Q011 MCQ

If $a > 3b$ and $c > d$, prove that $ac + 3bd > ad + 3bc$.

If $a > 3b$ and $c > d$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $(a - 3b)(c - d)$.
2. Factor the difference as $(a - 3b)(c - d)$.
3. The condition $a > 3b$ and $c > d$ makes each factor positive.
4. Hence $ac + 3bd > ad + 3bc$.

Correct option: A

Final answer

$$(a - 3b)(c - d)$$

Q012 Written

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

Solution

1. Subtract the right-hand side: $(3x - 4y) - (2x - 3y)$.
2. Simplify to $x - y$.
3. Since $x > y$, $x - y > 0$; therefore $3x - 4y > 2x - 3y$.

Final answer

$$3x - 4y > 2x - 3y$$

Worked Solutions

Inequalities

Q013 Written

Under the condition $x > 3$, $y > -5$, prove that $xy + 5x > 3y + 15$.

Solution

1. Start with LHS-RHS: $xy+5x-3y-15$.
2. Factor by grouping: $x(y+5) - 3(y+5) = (x-3)(y+5)$.
3. Since $x-3>0$ and $y+5>0$, the difference is positive; hence the inequality holds.

Final answer

$$xy + 5x > 3y + 15$$

Q014 Written

Under the condition $x > 2$, $y > 1$, prove that $xy + 2 > x + 2y$.

Solution

1. Start with LHS-RHS: $xy+2-x-2y$.
2. Factor: $x(y-1) - 2(y-1) = (x-2)(y-1)$.
3. Both factors are positive, so the difference is positive; hence $xy+2>x+2y$.

Final answer

$$xy + 2 > x + 2y$$

Worked Solutions

Inequalities

Q015 Written

Under the condition $a > 1$, $b > 1$, prove that $ab + 1 > a + b$.

Solution

1. Start with LHS-RHS: $ab+1-a-b$.
2. Factor: $ab-a-b+1=(a-1)(b-1)$.
3. Since $a-1>0$ and $b-1>0$, the difference is positive; hence $ab+1>a+b$.

Final answer

$$ab + 1 > a + b$$

Q016 Written

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

Solution

1. Subtract the right-hand side: $ax+by-ay-bx$.
2. Factor: $a(x - y) - b(x - y) = (a - b)(x - y)$.
3. Both factors are positive because $a>b$ and $x>y$; therefore the difference is positive.

Final answer

$$ax + by > ay + bx$$

Worked Solutions

Inequalities

Q017 Written

Under the condition $a > b > c > 0$, $x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

Solution

1. Subtract the right-hand side and regroup: $ax+by+cz-ay-bz-cx$.
2. Rewrite it as $(a-b)(x-y)+(b-c)(x-z)$.
3. Every factor in both products is positive, so the sum is positive; hence $ax+by+cz>ay+bz+cx$.

Final answer

$$ax + by + cz > ay + bz + cx$$

Worked Solutions

Inequalities

Q018 Challenge

If $a > b$ and $x > y$, prove that $2(ax + by) > 2(ay + bx)$.

If $a > b$ and $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $2(a - b)(x - y)$.
2. Factor the difference as $2(a - b)(x - y)$.
3. The condition $a > b$ and $x > y$ makes each factor positive.
4. Hence $2(ax + by) > 2(ay + bx)$.

Correct option: D

Final answer

$$2(a - b)(x - y)$$

Worked Solutions

Inequalities

Q019 Written

Prove that $a^2 + 4b^2 \geq 4ab$.**Solution**

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Complete the square: $a^2 + 4b^2 - 4ab = (a - 2b)^2 \geq 0$.
3. Therefore $a^2 + 4b^2 \geq 4ab$.
4. Equality holds when $a = 2b$.

Final answer

$$a^2 + 4b^2 \geq 4ab$$

Q020 MCQ

Under a,b real, prove that $a^2 + 4b^2 \geq 4ab$ **select the correct factorisation of LHS - RHS:****Solution**

1. Start with LHS - RHS: $a^2 + 4b^2 - 4ab$.
2. Factor/complete the square to obtain $(a - 2b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 4b^2 \geq 4ab$.

Correct option: D**Final answer**

$$(a - 2b)^2$$

Worked Solutions

Inequalities

Q021 MCQ

Under a, b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 9b^2 - 6ab$.
2. Factor/complete the square to obtain $(a - 3b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 9b^2 \geq 6ab$.

Correct option: C**Final answer**

$$(a - 3b)^2$$

Q022 MCQ

Under a, b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 16b^2 - 8ab$.
2. Factor/complete the square to obtain $(a - 4b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 16b^2 \geq 8ab$.

Correct option: B**Final answer**

$$(a - 4b)^2$$

Worked Solutions

Inequalities

Q023 MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2 - 10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$$(a - 5b)^2$$

Q024 MCQ

Under a, b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 36b^2 - 12ab$.
2. Factor/complete the square to obtain $(a - 6b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 36b^2 \geq 12ab$.

Correct option: C

Final answer

$$(a - 6b)^2$$

Worked Solutions

Inequalities

Q025 MCQ

Under a, b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 49b^2 - 14ab$.
2. Factor/complete the square to obtain $(a - 7b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 49b^2 \geq 14ab$.

Correct option: B**Final answer**

$$(a - 7b)^2$$

Q026 Written

Prove that $a^2 + 5b^2 \geq 4ab$.

Solution

1. Start with LHS - RHS: $a^2 + 5b^2 - 4ab$.
2. Complete the square: $a^2 + 5b^2 - 4ab = (a - 2b)^2 + b^2 \geq 0$.
3. Therefore $a^2 + 5b^2 \geq 4ab$.
4. Equality holds when $a = b = 0$.

Final answer

$$a^2 + 5b^2 \geq 4ab$$

Worked Solutions

Inequalities

Q027 Written

Prove that $9x^2 \geq y(6x - y)$.**Solution**

1. Subtract the right-hand side: $9x^2 - y(6x - y)$.
2. Complete the square: $9x^2 - y(6x - y) = (3x - y)^2 \geq 0$.
3. Therefore $9x^2 \geq y(6x - y)$.
4. Equality holds when $y = 3x$.

Final answer

$$9x^2 \geq y(6x - y)$$

Q028 Written

Prove that $2(x^2 + y^2) \geq (x + y)^2$.**Solution**

1. Subtract the right-hand side: $2(x^2 + y^2) - (x + y)^2$.
2. This is $(x - y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = y$.

Final answer

$$2(x^2 + y^2) \geq (x + y)^2$$

Worked Solutions

Inequalities

Q029 Written

Prove that $(2x - y)^2 \geq 3(x^2 - y^2)$.**Solution**

1. Subtract the right-hand side: $(2x - y)^2 - 3(x^2 - y^2)$.
2. Simplify to $(x - 2y)^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $x = 2y$.

Final answer

$$(2x - y)^2 \geq 3(x^2 - y^2)$$

Q030 Written

Prove that $(2a + 3b)^2 \geq 12ab$.**Solution**

1. Subtract the right-hand side: $(2a + 3b)^2 - 12ab$.
2. The cross terms cancel, giving $4a^2 + 9b^2 \geq 0$.
3. Therefore the inequality holds.
4. Equality holds when $a = b = 0$.

Final answer

$$(2a + 3b)^2 \geq 12ab$$

Worked Solutions

Inequalities

Q031 MCQ

Under x, y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $4x^2 - y(4x - y)$.
2. Factor/complete the square to obtain $(2x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $4x^2 \geq y(4x - y)$.

Correct option: A

Final answer

$$(2x - y)^2$$

Q032 MCQ

Under x, y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $9x^2 - y(6x - y)$.
2. Factor/complete the square to obtain $(3x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $9x^2 \geq y(6x - y)$.

Correct option: C

Final answer

$$(3x - y)^2$$

Worked Solutions

Inequalities

Q033 MCQ

Under x, y real, prove that $16x^2 \geq y(8x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $16x^2 - y(8x - y)$.
2. Factor/complete the square to obtain $(4x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $16x^2 \geq y(8x - y)$.

Correct option: D**Final answer**

$$(4x - y)^2$$

Q034 MCQ

Under x, y real, prove that $25x^2 \geq y(10x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $25x^2 - y(10x - y)$.
2. Factor/complete the square to obtain $(5x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $25x^2 \geq y(10x - y)$.

Correct option: D**Final answer**

$$(5x - y)^2$$

Worked Solutions

Inequalities

Q035 MCQ

Under x, y real, prove that $36x^2 \geq y(12x - y)$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $36x^2 - y(12x - y)$.
2. Factor/complete the square to obtain $(6x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $36x^2 \geq y(12x - y)$.

Correct option: A**Final answer**

$$(6x - y)^2$$

Q036 MCQ

Under x, y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Correct option: C**Final answer**

$$(x - y)^2$$

Worked Solutions

Inequalities

Q037 Written

Prove that $x^2 + y^2 \geq 2(x - y - 1)$.**Solution**

1. Start with LHS - RHS: $x^2 + y^2 - 2(x - y - 1)$.
2. Complete the squares: $x^2 + y^2 - 2(x - y - 1) = (x - 1)^2 + (y + 1)^2 \geq 0$.
3. Therefore $x^2 + y^2 \geq 2(x - y - 1)$.
4. Equality holds when $x = 1$, $y = -1$.

Final answer

$$x^2 + y^2 \geq 2(x - y - 1)$$

Q038 Written

Prove that $x^2 + 2 > 2x$.**Solution**

1. Subtract the right-hand side: $x^2 + 2 - 2x$.
2. Complete the square: $x^2 + 2 - 2x = (x - 1)^2 + 1 > 0$.
3. Therefore $x^2 + 2 > 2x$.
4. There is no equality case.

Final answer

$$x^2 + 2 > 2x$$

Worked Solutions

Inequalities

Q039 Written

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.**Solution**

1. Complete the squares: $x^2 + y^2 - 6x + 8y + 25 = (x - 3)^2 + (y + 4)^2 \geq 0$.
2. Therefore the inequality holds.
3. Equality holds when $x = 3$, $y = -4$.

Final answer

$$x^2 + y^2 - 6x + 8y + 25 \geq 0$$

Q040 Written

Prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.**Solution**

1. Complete the squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x - 5)^2 + 2(y + 3)^2 \geq 0$.
2. Therefore the expression is never negative.
3. Equality holds when $x = 5$, $y = -3$.

Final answer

$$x^2 + 2y^2 - 10x + 12y + 43 \geq 0$$

Worked Solutions

Inequalities

Q041 **Written****Prove that** $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$.**Solution**

1. Subtract the right-hand side: $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2$.
2. Use $(a^2 + b^2)(x^2 + y^2) - (ax + by)^2 = (ay - bx)^2 \geq 0$.
3. Therefore the required inequality holds.
4. Equality holds when $ay = bx$.

Final answer

$$(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$$

Worked Solutions

Inequalities

Q042 Challenge

For real x, y , prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Solution

1. Complete both squares: $x^2 + 2y^2 - 10x + 12y + 43 = (x - 5)^2 + 2(y + 3)^2 \geq 0$.
2. Therefore $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.
3. Equality holds when $x = 5$, $y = -3$.

Final answer

0

Worked Solutions

Inequalities

Q043 Written

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Solution

1. Both sides are positive. Compare their squares.

$$2. \left(\sqrt{\frac{a^2+b^2}{2}} \right)^2 - \left(\frac{a+b}{2} \right)^2 = \frac{(a-b)^2}{4} \geq 0.$$

3. Hence the required inequality holds.

4. Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$$

Worked Solutions

Inequalities

Q044 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-b)^2}{4}$$

Worked Solutions

Inequalities

Q045 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-2b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-2b)^2}{4}$$

Worked Solutions

Inequalities

Q046 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-3b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: C

Final answer

$$\frac{(a-3b)^2}{4}$$

Worked Solutions

Inequalities

Q047 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-4b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.
- Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-4b)^2}{4}$$

Worked Solutions

Inequalities

Q048 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-5b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-5b)^2}{4}$$

Worked Solutions

Inequalities

Q049 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

- Both sides are positive under the stated condition; compare their squares.
- The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-6b)^2}{4}$.
- This expression is non-negative under the stated conditions.
- Therefore $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.
- Equality occurs when the displayed square is zero.

Correct option: B

Final answer

$$\frac{(a-6b)^2}{4}$$

Worked Solutions

Inequalities

Q050 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-7b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions

Inequalities

Q051 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-8b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-8b)^2}{4}$$

Worked Solutions

Inequalities

Q052 Written

Prove that $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{2(a+b)})^2 - (\sqrt{a} + \sqrt{b})^2 = (\sqrt{a} - \sqrt{b})^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = b$.

Final answer

$$\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$$

Q053 Written

Prove that $a + 2 \geq \sqrt{3 + 6a}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(a + 2)^2 - (\sqrt{3 + 6a})^2 = (a - 1)^2 \geq 0$.
3. Hence the required inequality holds.
4. Equality holds when $a = 1$.

Final answer

$$a + 2 \geq \sqrt{3 + 6a}$$

Worked Solutions

Inequalities

Q054 Written

Prove that $\sqrt{2(a^2 + b^2)} \geq a + b$.

Solution

- Both sides are positive. Compare their squares.
- $(\sqrt{2(a^2 + b^2)})^2 - (a + b)^2 = (a - b)^2 \geq 0$.
- Hence the required inequality holds.
- Equality holds when $a = b$.

Final answer

$$\sqrt{2(a^2 + b^2)} \geq a + b$$

Q055 Written

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$.

Solution

- Both sides are positive. Compare their squares.
- $\left(\sqrt{\frac{a+b}{2}}\right)^2 - \left(\frac{\sqrt{a} + \sqrt{b}}{2}\right)^2 = \frac{(\sqrt{a} - \sqrt{b})^2}{4} \geq 0$.
- Hence the required inequality holds.
- Equality holds when $a = b$.

Final answer

$$\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$$

Worked Solutions

Inequalities

Q056 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 2\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

Correct option: B

Final answer

$$2\sqrt{a}\sqrt{b}$$

Q057 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 4\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

Correct option: A

Final answer

$$4\sqrt{a}\sqrt{b}$$

Worked Solutions

Inequalities

Q058 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 6\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

Correct option: C

Final answer

$$6\sqrt{a}\sqrt{b}$$

Q059 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 8\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Correct option: B

Final answer

$$8\sqrt{a}\sqrt{b}$$

Worked Solutions

Inequalities

Q060 Written

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$$

Q061 Written

Prove that $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + \sqrt{b})^2 - (\sqrt{4a + b})^2 = 4\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$$

Worked Solutions

Inequalities

Q062 Written

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(\sqrt{a} + \sqrt{b})^2 - (\sqrt{a+b})^2 = 2\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a} + \sqrt{b} > \sqrt{a+b}$$

Q063 Written

Prove that $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$.**Solution**

1. Both sides are positive. Compare their squares.
2. $(2\sqrt{a} + 3\sqrt{b})^2 - (\sqrt{4a+9b})^2 = 12\sqrt{ab} > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$2\sqrt{a} + 3\sqrt{b} > \sqrt{4a+9b}$$

Worked Solutions

Inequalities

Q064 Written

Prove that $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.

Solution

1. Both sides are positive because $a > b > 0$. Compare their squares.
2. $(\sqrt{a-b})^2 - (\sqrt{a} - \sqrt{b})^2 = 2\sqrt{b}(\sqrt{a} - \sqrt{b}) > 0$.
3. Hence the strict inequality holds.
4. There is no equality case.

Final answer

$$\sqrt{a-b} > \sqrt{a} - \sqrt{b}$$

Q065 Written

If $a > b > 0$, **prove that** $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.

Solution

1. Both sides are positive. Compare their squares.
2. $\left(\sqrt{a} - \frac{b}{2\sqrt{a}}\right)^2 - (\sqrt{a-b})^2 = \frac{b^2}{4a} > 0$.
3. Therefore $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.
4. There is no equality case.

Final answer

$$\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$$

Worked Solutions

Inequalities

Q066 Challenge

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Solution

1. Both sides are positive under $a, b > 0$; compare their squares.
2. $(\sqrt{a} + 4\sqrt{b})^2 - (\sqrt{a + 16b})^2 = 8\sqrt{ab} > 0$.
3. Therefore $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Final answer

$$\sqrt{a + 16b}$$

Worked Solutions

Inequalities

Q067 Written

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.**Solution**

1. By AM–GM, $\frac{b}{a} + \frac{a}{b} \geq 2\sqrt{\frac{b}{a} \cdot \frac{a}{b}} = 2$.
2. Equality holds when $a = b$.

Final answer

$$\frac{b}{a} + \frac{a}{b} \geq 2$$

Q068 Written

Prove that $\frac{b}{3a} + \frac{12a}{b} \geq 4$.**Solution**

1. Apply AM–GM to the two positive terms: $\frac{b}{3a} + \frac{12a}{b} \geq 2\sqrt{4} = 4$.
2. Equality holds when $b = 6a$.

Final answer

$$\frac{b}{3a} + \frac{12a}{b} \geq 4$$

Worked Solutions

Inequalities

Q069 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{a^2+b^2}{ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{a^2+b^2}{ab} - 2$.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{a^2+b^2}{ab} \geq 2$.

Correct option: A**Final answer**

$$\frac{(a-b)^2}{ab}$$

Q070 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{4a^2 + b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{4a^2 + b^2}{2ab} - 2$.
2. Factor/complete the square to obtain $\frac{(2a - b)^2}{2ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{4a^2 + b^2}{2ab} \geq 2$.

Correct option: C

Final answer

$$\frac{(2a - b)^2}{2ab}$$

Worked Solutions

Inequalities

Q071 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{9a^2 + b^2}{3ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{9a^2 + b^2}{3ab} - 2$.
2. Factor/complete the square to obtain $\frac{(3a - b)^2}{3ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{9a^2 + b^2}{3ab} \geq 2$.

Correct option: B**Final answer**

$$\frac{(3a - b)^2}{3ab}$$

Q072 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{16a^2 + b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{16a^2 + b^2}{4ab} - 2$.
2. Factor/complete the square to obtain $\frac{(4a - b)^2}{4ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{16a^2 + b^2}{4ab} \geq 2$.

Correct option: A

Final answer

$$\frac{(4a - b)^2}{4ab}$$

Worked Solutions

Inequalities

Q073 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{25a^2 + b^2}{5ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{25a^2 + b^2}{5ab} - 2$.
2. Factor/complete the square to obtain $\frac{(5a - b)^2}{5ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{25a^2 + b^2}{5ab} \geq 2$.

Correct option: D**Final answer**

$$\frac{(5a - b)^2}{5ab}$$

Q074 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{36a^2 + b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{36a^2 + b^2}{6ab} - 2$.
2. Factor/complete the square to obtain $\frac{(6a - b)^2}{6ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{36a^2 + b^2}{6ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(6a - b)^2}{6ab}$$

Worked Solutions

Inequalities

Q075 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{49a^2 + b^2}{7ab} \geq 2$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{49a^2 + b^2}{7ab} - 2$.
2. Factor/complete the square to obtain $\frac{(7a - b)^2}{7ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{49a^2 + b^2}{7ab} \geq 2$.

Correct option: C**Final answer**

$$\frac{(7a - b)^2}{7ab}$$

Q076 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{64a^2 + b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{64a^2 + b^2}{8ab} - 2$.
2. Factor/complete the square to obtain $\frac{(8a - b)^2}{8ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{64a^2 + b^2}{8ab} \geq 2$.

Correct option: D

Final answer

$$\frac{(8a - b)^2}{8ab}$$

Worked Solutions

Inequalities

Q077 Written

Prove that $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$.**Solution**

1. Expand: $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{a}{b} + \frac{b}{a} + 2$.
2. Use $\frac{a}{b} + \frac{b}{a} \geq 2$ to obtain the lower bound 4.
3. Equality holds when $a = b$.

Final answer

$$(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$$

Q078 Written

Prove that $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$.**Solution**

1. Expand to $x^2 + 5 + \frac{4}{x^2}$.
2. Since $x^2 + \frac{4}{x^2} \geq 4$, the product is at least 9.
3. Equality holds when $x = \sqrt{2}$.

Final answer

$$\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$$

Worked Solutions

Inequalities

Q079 Written

Prove that $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$.**Solution**

1. Expand to $10 + \frac{a}{b} + \frac{16b}{a}$.
2. AM–GM gives $\frac{a}{b} + \frac{16b}{a} \geq 8$, so the product is at least 18.
3. Equality holds when $a = 4b$.

Final answer

$$(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$$

Q080 Written

Prove that $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$.**Solution**

1. Expand to $ab + 6 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 12.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$$

Worked Solutions

Inequalities

Q081 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+b)^2}{ab} - 4$.
2. Factor/complete the square to obtain $\frac{(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+b)^2}{ab} \geq 4$.

Correct option: D**Final answer**

$$\frac{(a-b)^2}{ab}$$

Q082 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab} - 25$.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4(a-b)^2}{ab}$$

Worked Solutions

Inequalities

Q083 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+9b)(9a+b)}{ab} - 100$.
2. Factor/complete the square to obtain $\frac{9(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+9b)(9a+b)}{ab} \geq 100$.

Correct option: C**Final answer**

$$\frac{9(a-b)^2}{ab}$$

Q084 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+16b)(16a+b)}{ab} - 289$.
2. Factor/complete the square to obtain $\frac{16(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+16b)(16a+b)}{ab} \geq 289$.

Correct option: D

Final answer

$$\frac{16(a-b)^2}{ab}$$

Worked Solutions

Inequalities

Q085 Written

Prove that $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$.**Solution**

1. Expand to $ab + 10 + \frac{9}{ab}$.
2. AM–GM gives $ab + \frac{9}{ab} \geq 6$, so the product is at least 16.
3. Equality holds when $ab = 3$.

Final answer

$$\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$$

Q086 Written

Prove that $a + \frac{4}{a} \geq 4$.**Solution**

1. Apply AM–GM: $a + \frac{4}{a} \geq 2\sqrt{4} = 4$.
2. Equality holds when $a = 2$.

Final answer

$$a + \frac{4}{a} \geq 4$$

Worked Solutions

Inequalities

Q087 Written

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.**Solution**

1. Apply AM–GM to $3ab$ and $\frac{4}{ab}$: the sum is at least $2\sqrt{12} = 4\sqrt{3}$.
2. Equality holds when $ab = \frac{2}{\sqrt{3}}$.

Final answer

$$3ab + \frac{4}{ab} \geq 4\sqrt{3}$$

Q088 Written

Prove that $a + b + \frac{9}{a+b} \geq 6$.**Solution**

1. Let $t = a + b > 0$. AM–GM gives $t + \frac{9}{t} \geq 6$.
2. Equality holds when $a + b = 3$.

Final answer

$$a + b + \frac{9}{a+b} \geq 6$$

Worked Solutions

Inequalities

Q089 Written

A rectangle has perimeter 200 cm. Prove that its area $LW \leq 2500 \text{ cm}^2$.

Solution

1. From the perimeter, $L + W = 100$.
2. AM–GM gives $\frac{L+W}{2} \geq \sqrt{LW}$, so $50 \geq \sqrt{LW}$.
3. Therefore $LW \leq 2500$.
4. Equality holds when $L = W = 50$.

Final answer

$$LW \leq 2500$$

Worked Solutions

Inequalities

Q090 Challenge

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Solution

1. Put the terms over a positive denominator: $\frac{b}{a} + \frac{a}{b} - 2 = \frac{(a-b)^2}{ab} \geq 0$.
2. Hence $\frac{b}{a} + \frac{a}{b} \geq 2$.
3. Equality holds when $a = b$.

Final answer

2

Worked Solutions

Inequalities

Quiz MCQ 1 Concept Quiz · MCQ

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + y > 2(x + y)$.

Correct option: A

Final answer

$$x - y$$

Quiz MCQ 2 Concept Quiz · MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS - RHS:

Solution

1. Start with LHS - RHS: $a^2 + 25b^2 - 10ab$.
2. Factor/complete the square to obtain $(a - 5b)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $a^2 + 25b^2 \geq 10ab$.

Correct option: A

Final answer

$$(a - 5b)^2$$

Worked Solutions

Inequalities

Quiz MCQ 3 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = \frac{(a-7b)^2}{4}$.
3. This expression is non-negative under the stated conditions.
4. Therefore $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.
5. Equality occurs when the displayed square is zero.

Correct option: A

Final answer

$$\frac{(a-7b)^2}{4}$$

Worked Solutions

Inequalities

Quiz MCQ 4 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS – RHS:

Solution

1. Start with LHS - RHS: $\frac{(a+4b)(4a+b)}{ab} - 25$.
2. Factor/complete the square to obtain $\frac{4(a-b)^2}{ab}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{(a+4b)(4a+b)}{ab} \geq 25$.

Correct option: B

Final answer

$$\frac{4(a-b)^2}{ab}$$

Quiz WRITTEN 5 Concept Quiz · Written

If $x > y$, prove that $3x + 4y > 2x + 5y$.

Solution

1. Subtract the right-hand side: $x - y$.
2. Factor the difference as $x - y$.
3. The condition $x > y$ makes each factor positive.
4. Hence $3x + 4y > 2x + 5y$.

Final answer

$$2x + 5y$$

Worked Solutions

Inequalities

Quiz WRITTEN 6 Concept Quiz · Written

Under x, y real, prove that $x^2 + y^2 \geq 2xy$.

Solution

1. Start with LHS - RHS: $x^2 + y^2 - 2xy$.
2. Factor/complete the square to obtain $(x - y)^2$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $x^2 + y^2 \geq 2xy$.

Final answer

$$2xy$$

Quiz WRITTEN 7 Concept Quiz · Written

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Solution

1. Both sides are positive under the stated condition; compare their squares.
2. The difference of the squared sides is $\text{LHS}^2 - \text{RHS}^2 = 12\sqrt{a}\sqrt{b}$.
3. This expression is positive under the stated conditions.
4. Therefore $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Final answer

$$\sqrt{a + 36b}$$

Worked Solutions

Inequalities

Quiz WRITTEN 8 Concept Quiz · Written

Under $t > 0$, prove that $\frac{t^2+1}{t} \geq 2$.

Solution

1. Start with LHS - RHS: $\frac{t^2+1}{t} - 2$.
2. Factor/complete the square to obtain $\frac{(t-1)^2}{t}$.
3. Use the stated domain to show the factors are non-negative (strictly positive where required).
4. Hence $\frac{t^2+1}{t} \geq 2$.

Final answer

2

Answer key

Use the question number shown on each page.

Q001 $2(ax + by) > (a + b)(x + y)$

Q002 A

Q003 C

Q004 D

Q005 D

Q006 B

Q007 $ac + bd > ad + bc$

Q008 $ab + cd > ad + bc$

Q009 D

Q010 C

Q011 A

Q012 $3x - 4y > 2x - 3y$

Q013 $xy + 5x > 3y + 15$

Q014 $xy + 2 > x + 2y$

Q015 $ab + 1 > a + b$

Q016 $ax + by > ay + bx$

Q017 $ax + by + cz > ay + bz + cx$

Q018 D

Q019 $a^2 + 4b^2 \geq 4ab$

Q020 D

Q021 C

Q022 B

Q023 A

Q024 C

Q025 B

Q026 $a^2 + 5b^2 \geq 4ab$

Q027 $9x^2 \geq y(6x - y)$

Q028 $2(x^2 + y^2) \geq (x + y)^2$

Q029 $(2x - y)^2 \geq 3(x^2 - y^2)$

Q030 $(2a + 3b)^2 \geq 12ab$

Q031 A

Q032 C

Q033 D

Q034 D

Q035 A

Q036 C

Q037 $x^2 + y^2 \geq 2(x - y - 1)$

Q038 $x^2 + 2 > 2x$

Q039 $x^2 + y^2 - 6x + 8y + 25 \geq 0$

Q040 $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$

Q041 $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$

Q042 0

Q043 $\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2}$

Q044 A

Q045 A

Q046 C

Q047 A

Q048 A

Q049 B

Q050 A

Q051 A

Q052 $\sqrt{2(a + b)} \geq \sqrt{a} + \sqrt{b}$

Q053 $a + 2 \geq \sqrt{3 + 6a}$

Q054 $\sqrt{2(a^2 + b^2)} \geq a + b$

Q055 $\sqrt{\frac{a + b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$

Q056 B

Q057 A

Q058 C

Q059 B

Q060 $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$

Q061 $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$

Q062 $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$

Q063 $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$

Q064 $\sqrt{a - b} > \sqrt{a} - \sqrt{b}$

Q065 $\sqrt{a - b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$

Q066 $\sqrt{a + 16b}$

Q067 $\frac{b}{a} + \frac{a}{b} \geq 2$

Q068 $\frac{b}{3a} + \frac{12a}{b} \geq 4$

Q069 A

Q070 C

Q071 B

Q072 A

Q073 D

Q074 D

Q075 C

Q076 D

Q077 $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$

Q078 $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$

Q079 $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$

Q080 $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$

Q081 D

Q082 B

Q083 C

Q084 D

Q085 $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$

Q086 $a + \frac{4}{a} \geq 4$

Q087 $3ab + \frac{4}{ab} \geq 4\sqrt{3}$

Q088 $a + b + \frac{9}{a + b} \geq 6$

Q089 $LW \leq 2500$

Q090 2

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $2x + 5y$

Quiz WRITTEN 6 $2xy$

Quiz WRITTEN 7 $\sqrt{a + 36b}$

Quiz WRITTEN 8 2